THE BALANCED-PAIR ROUTE TO THE PISOT SUBSTITUTION CONJECTURE: FINITE OBSTRUCTIONS, TWO OPEN GATES, AND THE CURRENT FRONTIER

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Abstract. We give a self-contained account of the balanced-pair approach to the Pisot substitution conjecture for primitive substitutions with irreducible characteristic polynomial and Pisot dilation, without any unimodularity assumption. The approach reduces pure discrete spectrum of the associated tiling flow to two statements about the balanced-pair automaton $\mathcal{B}_\sigma$: finiteness of $\mathcal{B}_\sigma$ (G1) and productivity of every recurrent noncoincident strongly connected component (SCC Producer). We prove the reduction and then analyse both statements. On the finiteness side we show that unique decodability of the image code follows from $\det M_\sigma \neq 0$ alone, we record why it does not imply finiteness, and we isolate the remaining obligation as a renewal-finiteness statement (G1b-2) that is open; we prove a uniform finiteness theorem for multi-level cut ancestries under the Pisot hypothesis and identify, by explicit examples, which finite invariants of return words cannot be injective. On the coincidence side we prove a sink-SCC reduction, an exact intertwining $P_\mathcal{C} N_\mathcal{C} = M_\sigma P_\mathcal{C}$ forcing $\rho(N_\mathcal{C}) = \beta$ and $|\mathcal{C}| \geq |\mathcal{A}|$ for any finite closed nonproductive component, an unconditional endpoint-synchronization eliminator, a Barge–Diamond eliminator for three letters, a $\mathbb{Z}/2$ orientation cocycle with signed first-defect intertwiners $Q_r S = \Phi_r Q_r$, and spectral sieves on the first nonzero scattered-subword defect. We prove a wedge dichotomy for closed nonproductive components which shows that the "concentration" step of the spectral route is equivalent to the statement that no such component has vanishing degree-two defect, and that the remaining step of that route is equivalent to the statement that no such component has nonvanishing degree-two defect; both statements remain open, and together they are the coincidence problem itself. Exhaustive exact computations on all 4,554 primitive irreducible Pisot substitutions on three letters with image lengths at most three are reported as finite evidence with their exact hypotheses. The Pisot substitution conjecture remains open; G1 and G1b-2 remain open; no finite computation reported here is a finiteness theorem.

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1. Introduction

1.1. **The conjecture.** Let σ be a primitive substitution on a finite alphabet $\mathcal{A}$ with $|\mathcal{A}| = d \geq 2$, with incidence matrix M_σ. Suppose the characteristic polynomial χ_{M_σ} is irreducible over $\mathbb{Q}$ and its Perron–Frobenius eigenvalue $\beta > 1$ is a Pisot number. The *Pisot substitution conjecture* asserts that the substitution tiling flow $(\Omega_\sigma, \mathbb{R})$, with tile lengths given by a left Perron eigenvector, has pure discrete spectrum; equivalently (for irreducible Pisot substitutions) that the subshift (X_σ, S) has pure discrete spectrum. We refer to the survey of Akiyama, Barge, Berthé, Lee and Siegel [1] for the history, the equivalent formulations, and the state of the conjecture up to 2015.

The conjecture originates in Rauzy's construction of the Rauzy fractal [32] and the work of Arnoux and Ito [3] on strong coincidence. For two letters it is a theorem: Barge and Diamond [9] proved the strong coincidence condition, and Hollander and Solomyak [19] derived pure discrete spectrum, with Sirvent and Solomyak [34] relating the balanced-pair and overlap algorithms for the symbolic and the tiling actions. For larger alphabets the conjecture is known for several families: β-substitutions [7], families treated by Ito–Rao [20], Barge–Kwapisz [10], Akiyama–Lee [2], and Mercat–Akiyama [25]; large computer searches [5] have found no counterexample. A claimed general proof by Nakaishi [30] has not, to our knowledge, appeared in refereed form, and an earlier preprint by the same author was withdrawn [29]; we do not rely on either. Non-unimodular ("non-unit") Pisot substitutions require an internal space with a non-Archimedean factor [26, 4]; nothing below assumes $|\det M_\sigma| = 1$.

1.2. **Balanced pairs.** The balanced-pair algorithm goes back to Livshits [22] and was developed for Pisot substitutions by Sirvent and Solomyak [34]; see also Martensen [24] and [1, §5]. Starting from a pair of words with equal abelianization, one applies σ and cuts the inflated pair at every position where the two prefix abelianizations agree, producing a finite set of irreducible balanced pairs; the algorithm *terminates with coincidence* if the process generates only finitely many irreducible pairs and eventually produces a coincidence. For irreducible Pisot substitutions, termination with coincidence is equivalent to pure discrete spectrum [1, Thm. 5.3], and Barge, Štimac and Williams [11] give a form of the bridge that we use verbatim (Imported Theorem 2.16).

The balanced-pair formulation splits the conjecture into a *finiteness* statement and a *coincidence* statement about a directed graph, the balanced-pair automaton $\mathcal{B}_\sigma$ (Definition 2.11). This paper is an account of what is known about each half within a specific research program, with every statement labelled by its logical status.

1.3. **Status vocabulary.** Every numbered statement below carries one of the following tags.

- [*Theorem*]: proved in this paper under the stated hypotheses.
- [*Imported*]: a published theorem, cited with the exact statement used.
- [*Conditional*]: proved here from named hypotheses that are themselves open.
- [*Finite computation*]: an exact finite calculation with stated search domain and certificate meaning; it proves nothing beyond its domain.
- [*Empirical*]: a pattern observed in finite computations, not a proposition.
- [*Conjectural bridge*]: a reformulation or implication that is proposed, with the missing step identified.
- [*Open*]: a precisely stated statement that is neither proved nor refuted.

We are deliberately conservative about novelty. Several of the structural results below (unique decodability from the defect theorem, the Parikh intertwining, the sink-SCC reduction) are elementary once formulated, and closely related ideas appear in the literature on return words [17], on overlap coincidence [35, 2], and on the balanced-pair algorithm [34, 1]; we claim only the specific formulations and proofs given here.

1.4. **Explicit status of the two headline questions.**

- **The Pisot substitution conjecture is open.** Theorem 3.1 proves pure discrete spectrum only under the two hypotheses G1 (finiteness of $\mathcal{B}_\sigma$) and SCC Producer. Neither is proved for general $|\mathcal{A}| \geq 3$. Nothing in this paper reduces either hypothesis to a proved statement.
- **G1b-2 (renewal finiteness, Open Problem 4.5) is open.** It is the exact remaining obligation on the finiteness side once bounded discrepancy (Hypothesis 4.4, itself open) is granted. No bounded-norm argument can close it (Section 4.9); the exact computations of Section 6.1 show why, and prove nothing about it.
- **The strongest spectral route to SCC Producer contains no proved implication from wedge data to productivity.** We show (Proposition 5.20) that its "concentration" step is equivalent to excluding closed nonproductive components with $K_2 \equiv 0$ (Open Problem 5.22), and that its final step is equivalent to excluding those with $K_2 \not\equiv 0$ (Open Problem 5.23). Neither is proved; the higher-degree sieves of Section 5.5 constrain the first case but do not close it.

1.5. **Organisation.** Section 2 fixes hypotheses, proves unique decodability and aperiodicity, and defines the automaton and the bridge. Section 3 states the principal results. Sections 4 and 5 treat finiteness and coincidence respectively. Section 6 states the exact finite computations. Section 7 records counterexamples and retired routes. Section 8 lists the unresolved statements precisely, and Section 9 discusses what would close them.

2. Standing hypotheses and preliminaries

2.1. **Substitutions.** Let $\mathcal{A}$ be a finite alphabet, $|\mathcal{A}| = d$, and let $\mathcal{A}^*$ be the free monoid and $\mathcal{A}^+ = \mathcal{A}^* \setminus \{\varepsilon\}$. A *substitution* is a monoid morphism $\sigma : \mathcal{A}^* \to \mathcal{A}^*$ with $\sigma(a) \in \mathcal{A}^+$ for every a. The *Parikh map* $\pi : \mathcal{A}^* \to \mathbb{Z}^d$ sends w to its vector of letter counts; the *incidence matrix* $M = M_\sigma \in \mathrm{Mat}_d(\mathbb{Z}_{\geq 0})$ has $M_{ij} = |\sigma(j)|_i$, so that $\pi(\sigma(w)) = M\pi(w)$. For $w = w_1 \cdots w_n$ and $0 \leq k \leq n$ we write $w[k] = w_1 \cdots w_k$ for the prefix of length k.

Definition 2.1 (Standing hypotheses)**.** Throughout, unless explicitly stated otherwise, σ satisfies:

- (H1) M_σ is primitive: $M_\sigma^k > 0$ entrywise for some $k \geq 1$;
- (H2) χ_{M_σ} is irreducible over $\mathbb{Q}$, and $d = |\mathcal{A}| \geq 2$;
- (H3) the Perron–Frobenius eigenvalue β of M_σ is a Pisot number: $\beta > 1$ and every other root of χ_{M_σ} has modulus < 1.

We call such σ *PIP* (primitive, irreducible, Pisot). We do *not* assume $|\det M_\sigma| = 1$. The roots of χ_{M_σ} are written $\beta = \beta_1, \beta_2, \ldots, \beta_d$ with $1 > |\beta_2| \geq \cdots \geq |\beta_d| > 0$.

Lemma 2.2 ([*Theorem*])**.** *Under (H2),* $\det M_\sigma \neq 0$*, and* $\beta \notin \mathbb{Q}$*.*

Proof. If $\det M_\sigma = 0$ then $x \mid \chi_{M_\sigma}(x)$, contradicting irreducibility with $d \geq 2$. A rational root of an irreducible polynomial of degree ≥ 2 is impossible. ☐

Let $\ell \in \mathbb{R}^d_{>0}$ be a left Perron eigenvector, $\ell^\top M_\sigma = \beta \ell^\top$; its entries are the *tile lengths*, and $L(w) := \langle \ell, \pi(w) \rangle$ is the geometric length of w. Then $L(\sigma(w)) = \beta L(w)$.

Lemma 2.3 ($\mathbb{Z}$-independence of tile lengths; [*Theorem*])**.** *Under (H1)–(H2), the entries ℓ_a are linearly independent over* $\mathbb{Q}$*.*

Proof. If $n \in \mathbb{Z}^d \setminus \{0\}$ satisfies $\langle n, \ell \rangle = 0$, then $\langle (M^\top)^k n, \ell \rangle = \beta^k \langle n, \ell \rangle = 0$ for all k. Since $\chi_{M^\top} = \chi_M$ is irreducible, the minimal polynomial of $M^\top$ equals χ_M, so every nonzero vector is cyclic and $\{(M^\top)^k n\}_{k<d}$ spans $\mathbb{Q}^d$. Hence $\ell \perp \mathbb{Q}^d$, so $\ell = 0$, a contradiction. ☐

Let $X_\sigma \subseteq \mathcal{A}^\mathbb{Z}$ be the subshift generated by σ (the set of bi-infinite words all of whose factors are factors of some $\sigma^n(a)$), and let Ω_σ be the tiling space obtained by replacing letters by intervals of length ℓ_a; both carry the natural actions. Under primitivity, X_σ is minimal and uniquely ergodic [31].

Lemma 2.4 (Aperiodicity; [*Theorem*])**.** *Under (H1)–(H3), X_σ is infinite; equivalently, no element of X_σ is periodic.*

Proof. Suppose X_σ is finite. Primitivity gives a power $\tau = \sigma^q$ and a letter a with $\tau(a) \in a\mathcal{A}^*$; the one-sided fixed point $x = \lim_n \tau^n(a)$ has the same language as σ, so every factor of x occurs in one of finitely many periodic points; hence the factor complexity of x is bounded and x is eventually periodic by the Morse–Hedlund theorem [23, Thm. 1.3.13]; by uniform recurrence of x (primitivity) it is purely periodic, $x = w^\infty$ with w primitive (not a proper power). Then $\tau(w)^\infty = \tau(x) = x = w^\infty$, so $\tau(w)$ and w are powers of a common word [23, Prop. 1.3.4], and primitivity of w gives $\tau(w) = w^k$ for some integer $k \geq 1$. Applying π, $M^q \pi(w) = k\pi(w)$ with $\pi(w) \geq 0$ nonzero; since M^q is primitive, a nonnegative eigenvector belongs to the Perron eigenvalue, so $k = \beta^q$. Let $\alpha = \beta_2$ and let $\iota : \mathbb{Q}(\beta) \to \mathbb{C}$ be the field embedding with $\iota(\beta) = \alpha$ (it exists since χ_M is irreducible). Then $\alpha^q = \iota(\beta^q) = \iota(k) = k \geq 1$, contradicting $|\alpha| < 1$. ☐

Remark 2.5. Primitivity alone does not imply aperiodicity: $a \mapsto ab$, $b \mapsto ab$ is primitive with periodic subshift; its incidence matrix is singular and its characteristic polynomial is reducible. Lemma 2.4 is used only to justify the next imported theorem.

Imported Theorem 2.6 (Mossé recognizability [27, 28]; [*Imported*])**.** *Let σ be a primitive substitution whose subshift X_σ is aperiodic. There is $R = R(\sigma)$ such that for every $x \in X_\sigma$ the positions of the level-one supertile boundaries (the cutting points of a desubstitution $x = S^j\sigma(y)$, $y \in X_\sigma$) at a position i are determined by the factor $x_{[i-R,i+R]}$; the desubstitution is unique. Consequently every $x \in X_\sigma$ has a unique supertile hierarchy at every level.*

The Pisot hypothesis plays no role in Theorem 2.6; it enters through Lemma 2.4 and through contraction and spectral arguments below.

2.2. **Unique decodability.**

Imported Theorem 2.7 (Defect theorem [13, 14]; [*Imported*])**.** *Let $X \subset \mathcal{A}^+$ be finite. If X is not a code (i.e. some word admits two distinct factorizations into elements of X), then $X \subseteq H^*$ for a code $H \subset \mathcal{A}^+$ with $|H| \leq |X| - 1$.*

Theorem 2.8 (Unique decodability; [*Theorem*])**.** *Let σ be any substitution with* $\det M_\sigma \neq 0$*. Then $X_1 := \{\sigma(a) : a \in \mathcal{A}\}$ is a code with $|X_1| = d$, and for every $r \geq 1$ the set $\{\sigma^r(a) : a \in \mathcal{A}\}$ is a code.*

Proof. If $\sigma(a) = \sigma(b)$ with $a \neq b$ then two columns of M_σ coincide, so $\det M_\sigma = 0$; hence $|X_1| = d$. If X_1 were not a code, Theorem 2.7 gives a code H with $|H| \leq d-1$ and $X_1 \subseteq H^*$; writing $\sigma(a) = h_1 \cdots h_{k_a}$ and applying π, every column of M_σ lies in the $\mathbb{Z}$-span of the at most $d-1$ vectors $\pi(h)$, $h \in H$, so $\operatorname{rank} M_\sigma \leq d-1$, contradicting $\det M_\sigma \neq 0$. For powers, $\det M_{\sigma^r} = (\det M_\sigma)^r \neq 0$. □

Theorem 2.8 uses neither primitivity, nor aperiodicity, nor the Pisot property; it is strictly weaker than the recognizability results of [15], which cover morphisms of lower rank. Its only role below is to make hierarchies inside supertiles unique (Lemma 4.1) and to justify Remark 4.2.

2.3. **Balanced pairs and the automaton.**

Definition 2.9. A *balanced pair* is a pair $(u, v) \in \mathcal{A}^n \times \mathcal{A}^n$, $n \geq 1$, with $\pi(u) = \pi(v)$. Its *prefix-difference walk* is $D_{(u,v)}(k) = \pi(u[k]) - \pi(v[k]) \in \mathbb{Z}^d$, $0 \leq k \leq n$; a position k is a *zero return* if $D_{(u,v)}(k) = 0$. The pair is *irreducible* if its only zero returns are $k = 0$ and $k = n$. A *coincidence block* is an irreducible balanced pair with $u = v$; necessarily $n = 1$, so coincidence blocks are the pairs (a, a). Cutting a balanced pair at all its zero returns yields its *reduction* $\operatorname{red}(u, v) = (R_1, \ldots, R_m)$, an ordered sequence of irreducible balanced pairs (its *blocks*). We identify (u, v) with (v, u) when we speak of *states*; the *normalized representative* of a state is the pair whose first side is lexicographically smaller (the two sides differ unless the state is a coincidence block). We write $|T| = n$ for the length of a state T.

Lemma 2.10 ([*Theorem*])**.** *Let (u, v) be balanced and k a zero return. Then $|\sigma(u[k])| = |\sigma(v[k])| =:$ K and K is a zero return of $(\sigma(u), \sigma(v))$. Zero returns of $(\sigma(u), \sigma(v))$ of this form are called* inherited*; all others are* newborn*.*

Proof. $\pi(\sigma(u[k])) = M\pi(u[k]) = M\pi(v[k]) = \pi(\sigma(v[k]))$; equal Parikh vectors have equal lengths. □

Definition 2.11 (Balanced-pair automaton)**.** Let Seeds $= \{(ab, ba) : a, b \in \mathcal{A},\ a \neq b\}$ (modulo side swap). The *children* of a state $T = (u, v)$ are the states of the blocks of $\operatorname{red}(\sigma(u), \sigma(v))$, counted with multiplicity and order. The *balanced-pair automaton* $\mathcal{B}_\sigma$ is the directed graph whose vertex set is the set of states reachable from Seeds by iterating the child relation, with an edge $T \to U$ whenever U is a child of T; coincidence blocks are terminal (their children are not formed). A state is *productive* if some directed path from it reaches a coincidence block. A strongly connected component (SCC) of $\mathcal{B}_\sigma$ is *recurrent* if it contains a directed cycle (including loops), *noncoincident* if it contains no coincidence block, *closed* if every *noncoincident* child of every state in it lies in it (coincidence children are allowed), and *nonproductive* if none of its states is productive. A closed *noncoincident* SCC is nonproductive if and only if no state in it has a coincidence child (a coincidence block is itself a closed singleton SCC and is productive). The balanced-pair algorithm started at a seed s *terminates with coincidence* if the set of states reachable from s is finite and every one of them is productive.

Lemma 2.12 (Depth-n descendants; [*Theorem*])**.** *For every state $T = (u, v)$ and $n \geq 0$ the blocks of $\operatorname{red}(\sigma^n(u), \sigma^n(v))$ are exactly the states at the ends of the directed paths of length n from T in the child graph (with multiplicity), and the boundaries between them are exactly the zero returns of $(\sigma^n(u), \sigma^n(v))$.*

Proof. Induction on n using Lemma 2.10: the zero returns of $(\sigma^n(u), \sigma^n(v))$ consist of the inherited images of the zero returns of level $n-1$, which partition the pair into the inflations of the level-$(n-1)$ blocks, together with the interior zero returns of each such inflated block; the latter define exactly the children of that block. Swapping the two sides of a block swaps the sides of all its children and does not change the states. □

Remark 2.13 (Seeds and legality)*.* The seed words ab, ba need not be factors of the language of σ (for the Tribonacci substitution $1 \mapsto 12, 2 \mapsto 13, 3 \mapsto 1$ the word 23 is not a factor). Consequently $\mathcal{B}_\sigma$ may contain states that are not realized by pairs of points of X_σ; we return to this in Section 5.9. Using all seeds, rather than a single one, is what allows the graph-theoretic reductions of Section 5; the price is the seed-family question of Section 4.8.

Hypothesis 2.14 (G1, finiteness; [*Open*])**.** *$\mathcal{B}_\sigma$ is finite.*

Conjecture 2.15 (SCC Producer; [*Open*])**.** *Every recurrent noncoincident SCC of $\mathcal{B}_\sigma$ contains a productive state.*

Since an SCC is strongly connected, if one of its states is productive then all are. The conjecture is a component-level statement: individual states with no coincidence child are common (Section 6), so no state-by-state argument can prove it.

2.4. **The spectral bridge.**

Imported Theorem 2.16 ([*Imported*])**.** *Let σ be a primitive substitution with irreducible characteristic polynomial and Pisot dilation. If for some $a \neq b$ the balanced-pair algorithm started at (ab, ba) terminates with coincidence, then the tiling flow $(\Omega_\sigma, \mathbb{R})$ has pure discrete spectrum. This is the one-dimensional statement of Barge–Štimac–Williams* [11, Cor. 2]*; the equivalence between termination with coincidence and pure discrete spectrum for irreducible Pisot substitutions, with the seed clause, is recorded in* [1, Thm. 5.3]*, building on* [34, 19]*. For two letters, termination with coincidence follows from* [9, 19]*.*

We use only the forward implication stated. The equivalence between pure discrete spectrum of the flow and of the subshift for irreducible Pisot substitutions is discussed in [16, 1]; the earlier form of the bridge in [10, Prop. 17.4] depends on [10, Thm. 16.3], whose proof is reported in [11, §1] to contain a gap; we therefore cite [11].

3. Principal statements

Theorem 3.1 (Conditional main theorem; [*Conditional*])**.** *Let σ be PIP. If $\mathcal{B}_\sigma$ is finite (Hypothesis 2.14) and Conjecture 2.15 holds for σ, then $(\Omega_\sigma, \mathbb{R})$ has pure discrete spectrum.*

Proof. Fix $a \neq b$. The forward orbit of the seed (ab, ba) in the finite graph $\mathcal{B}_\sigma$ reaches only finitely many states, and every infinite forward path enters a recurrent SCC. If that SCC contains a coincidence block, the corresponding state is productive. Otherwise Conjecture 2.15 gives a productive state in it, and strong connectedness makes every state of the SCC productive. Since every state reachable from the seed either is a coincidence block, or leads to a recurrent SCC, or lies in one, every reachable state is productive, i.e. the algorithm from (ab, ba) terminates with coincidence. Imported Theorem 2.16 concludes. □

The rest of the paper concerns the two hypotheses. The following statements summarise what is proved about each; proofs are in Sections 4 and 5.

Theorem 3.2 (Sink reduction; [*Conditional*] on G1)**.** *If $\mathcal{B}_\sigma$ is finite and some state is nonproductive, then $\mathcal{B}_\sigma$ contains a finite closed nonproductive recurrent noncoincident SCC.*

Theorem 3.3 (Structure of a finite closed nonproductive component; [*Theorem*])**.** *Let σ be PIP and $\mathcal{C}$ a finite closed nonproductive recurrent noncoincident SCC. Let $P_\mathcal{C} \in \mathrm{Mat}_{d\times|\mathcal{C}|}(\mathbb{Z}_{\geq 0})$ have the common Parikh vectors of the states as columns, and let $N_\mathcal{C}$ count children. Then $P_\mathcal{C} N_\mathcal{C} = M_\sigma P_\mathcal{C}$, $\mathrm{rank}_\mathbb{Q} P_\mathcal{C} = d$, $|\mathcal{C}| \geq d$, $\chi_{M_\sigma} \mid \chi_{N_\mathcal{C}}$, and $\rho(N_\mathcal{C}) = \beta$. Moreover neither endpoint map $\sigma_\pm$ (first, resp. last, letter of $\sigma(a)$) is globally synchronizing, and for $d = 3$ neither endpoint map is a 3-cycle.*

Theorem 3.4 (Signed defect intertwiners; [*Theorem*])**.** *With $\mathcal{C}$ as above, child occurrences carry a $\mathbb{Z}/2$ orientation cocycle; if A, B count positively and negatively oriented occurrences then $N_\mathcal{C} = A + B$, $S := A - B$ satisfies $\rho(S) \leq \beta$, and at the first degree r at which some scattered-subword defect K_r is nonzero on $\mathcal{C}$, $Q_r S = \Phi_r Q_r$ where Q_r has the $K_r(T)$ as columns and Φ_r is the action of $M^{\otimes r}$ on the free Lie component $\mathrm{Lie}_r(\mathbb{Q}^d)$. For $d = 3$: if $r = 2$ then $\mathrm{rank}\, Q_2 = 3$ and $\chi_{\Lambda^2 M} \mid \chi_S$; if $r = 3$ then $\mathrm{im}\, Q_3 \subseteq \ker(\Phi_3 - \det M)$; if $r = 4$ then $|\det M| = 1$ and $\rho(S) = \beta$.*

Theorem 3.5 (Uniform return and ancestry finiteness; [*Theorem*])**.** *Let σ be PIP and $\mathcal{C}$ a finite set of noncoincident irreducible states closed under taking children. Then zero returns of $(\sigma^n(u), \sigma^n(v))$, $(u, v) \in \mathcal{C}$, occur with gaps at most $B_\mathcal{C} := \max_{T\in\mathcal{C}} |T|$ for every n, and the multi-level Parikh-defect ancestries of all such zero returns take values in one finite subset of $\mathbb{Z}^d$ independent of n.*

Proposition 3.6 (Wedge dichotomy, $d = 3$; [*Theorem*])**.** *Let $\mathcal{C}$ be a closed nonproductive SCC of $\mathcal{B}_\sigma$ (finite or not). Then either $K_2 \equiv 0$ on $\mathcal{C}$, or the vectors $K_2(T)$, $T \in \mathcal{C}$, span $\Lambda^2\mathbb{Q}^3$; in the second case every nonzero linear functional on $\Lambda^2\mathbb{R}^3$, in particular the dominant spectral projection of $\Lambda^2 M$, is nonzero at some $K_2(T)$.*

4. Level 2: finiteness of the automaton

4.1. What unique decodability does and does not give.

Lemma 4.1 (Unique hierarchy inside supertiles; [*Theorem*])**.** *Let $\det M_\sigma \neq 0$. For $\tau \in \mathcal{A}$ and $1 \leq k \leq n$, the level-k supertile boundaries inside $\sigma^n(\tau)$ are determined by the word $\sigma^n(\tau)$ alone.*

Proof. $\sigma^n(\tau) = \sigma^k(\sigma^{n-k}(\tau))$ is a concatenation of elements of the code $\{\sigma^k(a)\}$ (Theorem 2.8), whose factorization is unique. □

Remark 4.2 (Unique decodability does not imply finiteness). An earlier version of this program claimed that Theorem 2.8 bounds the total coincidence "padding" inside an inflated balanced pair and hence that the geometric length of a child s of a state s' satisfies $\beta L(s') \leq L(s) + D$ for a constant D, from which backward contraction would give finiteness. *Both claims are withdrawn.* Unique decodability excludes only coincidence runs whose phase pattern is a legally repeatable complete-cutting cycle; runs recurring at a nonzero within-image offset are not excluded, and long coincidence runs do occur. More decisively, inflation of s' may split into many noncoincident children none of which carries most of $\beta L(s')$; on the corpus of Section 6 the ratio $\beta L(s')/L(s)$ over child edges exceeds 8 and grows with state length. No form of this argument is used below.

4.2. Discrepancy and the two-gate decomposition.

Definition 4.3. The *discrepancy* of a balanced pair $s = (u, v)$ of length n is $\mathrm{Disc}(s) := \max_{0\le k\le n} \|D_s(k)\|_\infty$. A state is *reachable* if it is a vertex of $\mathcal{B}_\sigma$.

Hypothesis 4.4 (G1b-1, bounded discrepancy; [*Open*])**.** $\sup\{\mathrm{Disc}(T) : T \textit{ reachable}\} < \infty$.

Open Problem 4.5 (G1b-2, renewal finiteness; [*Open*])**.** *For every R_0, prove that only finitely many reachable irreducible states T satisfy* $\mathrm{Disc}(T) \le R_0$.

Lemma 4.6 ([*Conditional*])**.** *Hypothesis 4.4 together with a positive solution of Open Problem 4.5 implies Hypothesis 2.14.*

Proof. Take R_0 to be the supremum in Hypothesis 4.4. □

A contraction estimate of the form $\mathrm{Disc}(\sigma\text{-child}) \le c\, \mathrm{Disc}(\text{parent}) + E$ with $c < 1$, in a norm adapted to M_σ, has been asserted in unpublished working notes of this program; we have not been able to reconstruct a proof, and we treat Hypothesis 4.4 as open. We stress that even granting it, Open Problem 4.5 is a separate statement: the prefix-difference walk of an irreducible balanced pair is a first return to 0 of a walk in $\mathbb{Z}^d$ that avoids 0, and a bounded box contains arbitrarily long such walks. Finiteness must come from the substitutive structure of which walks actually occur.

4.3. Labelled first returns and the information-loss example.

For an irreducible balanced pair (u, v) the sequence $(D_{(u,v)}(k))_{0\le k\le n}$ is a first-return path of a walk with steps $e_{u_k} - e_{v_k}$. The path does not determine the pair: a diagonal step (a, a) contributes 0 for every a.

Example 4.7 ([*Finite computation*])**.** Over $\mathcal{A} = \{0, 1, 2\}$ the irreducible balanced pairs $(001, 100)$ and $(021, 120)$ have the same prefix-difference path $0 \to (1, -1, 0) \to (1, -1, 0) \to 0$; their middle steps are $(0, 0)$ and $(2, 2)$.

Hence any approach to Open Problem 4.5 that quotients states to their difference paths loses information; the labelled step sequence must be retained. The following proposition shows that, conversely, retaining bounded windows of substitutive ancestry data is also insufficient, for a specific non-unimodular substitution.

Example 4.8. Let $\tau : 0 \mapsto 1,\ 1 \mapsto 021,\ 2 \mapsto 001$ on $\mathcal{A} = \{0, 1, 2\}$. Its incidence matrix is

$$M_\tau = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \chi_{M_\tau}(x) = x^3 - x^2 - 2x - 2, \qquad \det M_\tau = 2.$$

One checks that $M_\tau^3 > 0$, that χ_{M_τ} has no rational root (its possible roots $\pm 1, \pm 2$ give values $-4, -2, -2, -10$), and that its real root $\beta \approx 2.27$ is Pisot: the remaining roots have product $2/\beta < 1$ and, being complex conjugates (the discriminant of χ is $-152 < 0$), modulus $\sqrt{2/\beta} < 1$. So τ is PIP and non-unimodular.

Proposition 4.9 (A regular family of zero returns; [*Theorem*])**.** *For τ as in Example 4.8 and $d \ge 2$, put $U_d = \tau^d(01)$, $V_d = \tau^d(10)$, $L_d = |U_d|$. Both U_d and V_d end with* 021*, and the three positions $L_d - 3, L_d - 2, L_d - 1$ are interior zero returns of (U_d, V_d). At each of these positions the two cuts lie, on both sides, inside the image of the source letter at index* 1 *of the seed (letter* 1 *on top, letter* 0 *below), with source prefix defect $(1, -1, 0)$, and the descent through the substitution hierarchy from level d to level* 0 *has, on the top side, digit sequence $(1, 2)^{d-1}(1, j)$ and, on the bottom side, $(0, 0)(1, 2)^{d-2}(1, j)$, $j \in \{0, 1, 2\}$, where a digit (a, i) records the source letter and the index of the child image containing the cut.*

Proof. $\tau^d(1)$ ends with $\tau(\text{last letter of } \tau^{d-1}(1)) = \tau(1) = 021$ for $d \geq 2$, and $\tau^d(0) = \tau^{d-1}(1)$ ends with 021 for $d \geq 3$, while $\tau^2(0) = 021$. Removing the common suffixes 021, 21, 1 preserves Parikh equality, giving the three zero returns; they are interior since $L_d > 3$. The remaining assertions are a direct descent through the images: the cut sits in the last child image of $\tau(1) = 021$ (letter 1, index 2) at each level above the bottom one, and on the bottom side the seed letter 0 contributes one initial digit $(0, 0)$ because $\tau(0) = 1$. □

Corollary 4.10 ([*Theorem*])**.** *For every source-context radius r and every digit-window size w, the map sending a certified interior zero return of (U_d, V_d) to the tuple (relative source displacement, source prefix defect, source letters, radius-r source context on both sides, first w and last w digits on both sides) is not injective on $\bigcup_{d\geq 2}\{L_d - 3, L_d - 2, L_d - 1\}$: the cuts at depths $d - 1$ and d with the same j collide as soon as $w \leq d - 2$.*

Thus the level-scaled address of a zero return (its full digit sequence) cannot be replaced by bounded end windows. What Proposition 4.9 also shows is that the offending family is *regular*: it is generated by a loop $(1, 2)^*$ in the digit alphabet. Whether every family of reachable states is regular in this sense, uniformly, is the content of Open Problem 4.15.

4.4. **Multi-level ancestry and Pisot finiteness.** Fix a balanced pair $T = (u, v)$ (not necessarily irreducible) and an output position t in $(\sigma(u), \sigma(v))$. On the top side let i be the number of complete images $\sigma(u_1) \cdots \sigma(u_i)$ preceding t and $r = t - |\sigma(u[i])|$ the offset inside $\sigma(u_{i+1})$ $(0 \leq r < |\sigma(u_{i+1})|$, with the convention $(i, r) = (|u|, 0)$ at the right end); define (j, s) on the bottom side likewise. Let $q_\sigma(a, r) = \pi(\sigma(a)[r])$.

Lemma 4.11 (Prefix-difference inflation; [*Theorem*])**.** *$D_{(\sigma(u),\sigma(v))}(t) = M\big(\pi(u[i]) - \pi(v[j])\big) + q_\sigma(u_{i+1}, r) - q_\sigma(v_{j+1}, s)$.*

Proof. The top prefix of length t is $\sigma(u[i])\,\sigma(u_{i+1})[r]$, with Parikh vector $M\pi(u[i]) + q_\sigma(u_{i+1}, r)$; similarly below. □

Write $E = \pi(u[i]) - \pi(v[j])$ (the *source defect*) and $c = q_\sigma(u_{i+1}, r) - q_\sigma(v_{j+1}, s)$ (the *correction digit*); the set C_σ of possible corrections is finite and depends only on σ. If t is a zero return then $ME + c = 0$. Since M is invertible over $\mathbb{Q}$, $E = -M^{-1}c$ is determined by c; in particular:

Corollary 4.12 (Finite one-step ancestry; [*Theorem*])**.** *The source defects below one-step zero returns lie in a finite set $A_1(\sigma) = \{-M^{-1}c : c \in C_\sigma\} \cap \mathbb{Z}^d$, and the source-index misalignment $i - j = \langle \mathbf{1}, E\rangle$ is bounded by a constant depending only on σ. An interior zero return of an irreducible pair cannot have $r = s = 0$: otherwise $c = 0$, $E = 0$, and $i = j$ would be an interior zero return of the parent.*

Iterating the descent gives, for a zero return t at level n (in $(\sigma^n(u), \sigma^n(v))$), a sequence of source defects $x_n = 0, x_{n-1}, \ldots, x_0$ with $x_{k+1} = Mx_k + c_{k+1}$, $c_{k+1} \in C_\sigma$, where x_0 is a difference of prefix Parikh vectors of (u, v) at possibly different positions. Expanding, $M^n x_0 + \sum_k M^{n-1-k} c_{k+1} = 0$; this identity is the *level-scaled address certificate* of the zero return, and it uses no inverse of M and no choice of completion.

Theorem 4.13 (Pisot ancestry finiteness; [*Theorem*])**.** *Let σ be PIP, and let $F, D \subset \mathbb{Z}^d$ be finite. There is a finite set $A = A(M, F, D) \subset \mathbb{Z}^d$ containing every term of every sequence $x_0, \ldots, x_n \in \mathbb{Z}^d$ (any n) with $x_0 \in F$, $x_n = 0$, and $x_{k+1} = Mx_k + d_{k+1}$, $d_{k+1} \in D$.*

Proof. Let $\mathbb{R}^d = E_u \oplus E_s$ where E_u is the Perron eigenline and E_s the M-invariant real complement spanned by the generalized eigenspaces of $\beta_2, \ldots, \beta_d$; $\rho(M|_{E_s}) < 1$, so $\|M^m \pi_s z\| \leq Kq^m \|\pi_s z\|$ for constants K, $0 < q < 1$. Forward iteration gives $\pi_s x_k = M^k \pi_s x_0 + \sum_{j\leq k} M^{k-j} \pi_s d_j$, bounded uniformly in k by $K(\max_F \|\pi_s\| + \max_D \|\pi_s\|/(1-q))$. On E_u the recurrence is $\pi_u x_{k+1} = \beta \pi_u x_k +$

$\pi_u d_{k+1}$; solving backward from $x_n = 0$, $\pi_u x_k = -\sum_{j>k} \beta^{k-j}\pi_u d_j$, bounded by $\max_D \|\pi_u\|/(\beta-1)$. So all x_k lie in a fixed bounded region of $\mathbb{R}^d$, which meets $\mathbb{Z}^d$ in a finite set. □

Remark 4.14. Theorem 4.13 is the matrix-radix analogue of the finiteness of digit expansions in a Pisot base; the proof is self-contained. It does not assume unimodularity and does not treat any projection of $\mathbb{Z}^d$ as a lattice: only discreteness of $\mathbb{Z}^d$ itself is used. A finite ancestry alphabet does *not* by itself constrain the digit sequences: nonzero cycles occur (Example 7.3).

4.5. **Affine traces and the pump question.** Retaining, at each level k of a descent, the pair of source letters (a_k, b_k) containing the two cuts together with x_k gives the *affine trace* $\big((a_k, b_k, x_k)\big)_{0\le k\le n}$ of a zero return. For a fixed PIP σ and a fixed finite closed set of states, the affine states range over a finite set (Theorem 4.13), so every sufficiently deep zero return has a repeated affine state, i.e. a *loop* in its trace.

Open Problem 4.15 (Affine pump normalization; [*Open*])**.** *Prove (or refute): for a PIP substitution and a bound R_0, every reachable state with* Disc $\le R_0$ *whose ancestry trace contains a loop can be shortened along the loop to another reachable state with the same labelled local data, and the loop-free traces form a finite set. Either clause, proved uniformly, would resolve Open Problem 4.5 in the presence of Hypothesis 4.4.*

The finite computations of Section 6.1 show that, for τ of Example 4.8, every collision of a natural finite projection observed through depth seven is explained by such a loop. That is evidence for the shape of the statement, not for its truth.

4.6. **Design constraints for the non-unimodular case.** Any contracting-space argument for Open Problem 4.5 must respect the following facts. For unimodular PIP substitutions the projection of $\mathbb{Z}^d$ to the contracting hyperplane E_s is dense, not a lattice, so no naive discreteness argument in E_s is available. For non-unimodular substitutions the natural internal space is the product of E_s with a non-Archimedean factor $\prod_{p|\det M} K_p$ determined by the primes dividing the norm of β [26, 4]; a level-scaled address must retain the corresponding residue information. For τ of Example 4.8, $\det M_\tau = 2$ and $\beta(\beta^2-\beta-2) = 2$, so the completion of $\mathbb{Q}(\beta)$ at the prime above β is $\mathbb{Q}_2$, and the finite quotients $\mathbb{Z}^3/M_\tau^k\mathbb{Z}^3$ (of order 2^k) are the natural finite approximations of that factor. The computations of Section 6.1 use exactly these quotients and no floating-point or inverse-matrix arithmetic.

4.7. **Geometric reformulation and the overlap algorithm.** The overlap-coincidence algorithm of Solomyak [35] and Akiyama–Lee [2] decides pure discrete spectrum for self-affine tilings with the Meyer property by iterating inflation on the finite set of *realized* overlaps of two tilings in one fibre of the maximal equicontinuous factor; finiteness of that set follows from finite local complexity and the Meyer property, which holds for Pisot-family self-affine tilings [21]. In one dimension the balanced-pair and overlap algorithms are related in [34]. A reachable irreducible state (u, v) that is realized by two tilings $\mathcal{T}, \mathcal{T}'$ in one fibre corresponds to a maximal interval between consecutive simultaneous tile boundaries, and hence to a finite chain of overlaps.

Conjecture 4.16 (Relative density of simultaneous boundaries; [*Conjectural bridge*])**.** *For PIP σ and any two tilings $\mathcal{T}, \mathcal{T}' \in \Omega_\sigma$ in the same fibre of the maximal equicontinuous factor, the set of simultaneous boundaries $\partial\mathcal{T} \cap \partial\mathcal{T}'$ is relatively dense in $\mathbb{R}$, with a gap bound uniform in the fibre.*

A uniform gap bound would bound the length of every *realized* irreducible state, giving finiteness of the realized part of $\mathcal{B}_\sigma$; combined with a proof that every reachable state is realized (Section 5.9) it would give G1. We record this as the most natural geometric target for Open Problem 4.5; we do not claim it, and we note that it is not implied by the Meyer property alone, which controls overlaps of tiles but not gaps between coincidences of boundaries.

4.8. **The seed-family bridge.**

Proposition 4.17 (Seed union; [*Theorem*])**.** *Let $F_{a,b}$ be the set of states reachable from the single seed (ab, ba). Then the vertex set of $\mathcal{B}_\sigma$ is $\bigcup_{a\neq b} F_{a,b}$. In particular, if every single-seed algorithm terminates with coincidence then $\mathcal{B}_\sigma$ is finite and every state is productive.*

Proof. Immediate from Definition 2.11. □

Open Problem 4.18 ([*Open*])**.** *Determine whether, for irreducible Pisot σ with pure discrete spectrum, the balanced-pair algorithm terminates with coincidence from* every *seed (ab, ba), $a \neq b$, including seeds that are not legal factors.*

The statement in [1, Thm. 5.3] is that termination with coincidence is equivalent to pure discrete spectrum and that a seed of the form (ij, ji) suffices; we have not verified from the proofs in [34, 19] that the equivalence holds for an arbitrary chosen seed, nor for seeds outside the language. Until this is settled we do not assert "pure discrete spectrum implies G1" for the all-seed automaton. If the seedwise implication holds, then G1 is a *necessary* condition for pure discrete spectrum and Theorem 3.1 is not merely one sufficient route.

4.9. **Retired finiteness arguments.** The following are not valid routes to G1 and are recorded so that they are not reattempted: (i) unique decodability implies bounded total coincidence padding (Remark 4.2); (ii) bounded discrepancy implies finitely many states; (iii) the zero-sum hyperplane contracts under M in a fixed norm (false: the Perron component of a nonzero difference vector expands); (iv) every long state contains a uniformly short irreducible core; (v) the two-letter case is a trivial one-dimensional walk (the known two-letter proof [9, 19] uses genuine substitutive structure); (vi) the unlabelled difference path determines the state (Example 4.7).

5. Level 3: coincidence

Throughout this section σ is PIP unless a statement says otherwise.

5.1. **Sink-SCC reduction.**

Theorem 5.1 ([*Conditional*] on G1; Theorem 3.2)**.** *Let $\mathcal{B}$ be a finite balanced-pair automaton and* NP *its set of nonproductive states. If* NP $\neq \emptyset$*, then $\mathcal{B}$ contains a closed nonproductive recurrent noncoincident SCC. Consequently Conjecture 2.15 for a finite $\mathcal{B}_\sigma$ is equivalent to the nonexistence of a closed nonproductive recurrent noncoincident SCC.*

Proof. NP is forward closed: a child of a nonproductive state is nonproductive. The condensation of the induced graph $\mathcal{B}[\mathrm{NP}]$ is a finite nonempty acyclic digraph and has a sink vertex, an SCC $\mathcal{C}$. Every child of a state of $\mathcal{C}$ lies in NP and, $\mathcal{C}$ being a sink of the condensation of $\mathcal{B}[\mathrm{NP}]$, in $\mathcal{C}$; so $\mathcal{C}$ is closed in $\mathcal{B}$. Every noncoincident state has at least one child (the reduction of $(\sigma(u), \sigma(v))$ is nonempty), so an infinite forward path exists inside the finite set $\mathcal{C}$, which therefore contains a cycle. $\mathcal{C}$ contains no coincidence block since those are productive. Conversely a closed nonproductive SCC is a set of nonproductive states. □

Thus, under G1, it suffices to rule out one kind of object: a finite closed nonproductive recurrent noncoincident SCC. We call such an SCC a *strict component*. Sections 5.2–5.8 are a list of properties that every strict component must have.

5.2. **Mass balance and the Parikh intertwiner.** Let $\mathcal{C} = \{T_1, \dots, T_m\}$ be a finite closed nonproductive SCC. Nonproductivity and closure mean that every block of $\mathrm{red}(\sigma(u_j), \sigma(v_j))$ is a state of $\mathcal{C}$. Hence $\tau_{\mathcal{C}}(T_j) :=$ (ordered sequence of the states of those blocks) is a substitution on the alphabet $\mathcal{C}$, the *derived substitution*; its incidence matrix $N_{\mathcal{C}}$, $(N_{\mathcal{C}})_{ij} = |\tau_{\mathcal{C}}(T_j)|_{T_i}$, is irreducible because $\mathcal{C}$ is strongly connected. Let $p_j = \pi(u_j) = \pi(v_j)$ and $P_{\mathcal{C}} = [p_1 \cdots p_m] \in \mathrm{Mat}_{d\times m}(\mathbb{Z}_{\geq 0})$.

Theorem 5.2 ([*Theorem*]; part of Theorem 3.3). *$P_{\mathcal{C}}N_{\mathcal{C}} = MP_{\mathcal{C}}$; $\operatorname{rank}_{\mathbb{Q}} P_{\mathcal{C}} = d$ and $m \geq d$; $\chi_M \mid \chi_{N_{\mathcal{C}}}$ in $\mathbb{Z}[x]$; and $\rho(N_{\mathcal{C}}) = \beta$. If $m = d$ then $N_{\mathcal{C}} = P_{\mathcal{C}}^{-1}MP_{\mathcal{C}}$.*

Proof. The j-th column of $P_{\mathcal{C}}N_{\mathcal{C}}$ is the sum of the Parikh vectors of the blocks of $\operatorname{red}(\sigma(u_j), \sigma(v_j))$, which is $\pi(\sigma(u_j)) = Mp_j$. Let $W = \operatorname{im}_{\mathbb{Q}} P_{\mathcal{C}} \neq 0$; then $MW = \operatorname{im}(P_{\mathcal{C}}N_{\mathcal{C}}) \subseteq W$. The minimal polynomial of $M|_W$ divides the irreducible χ_M, hence equals it, so $\dim W \geq d$, i.e. $W = \mathbb{Q}^d$. Since $P_{\mathcal{C}} : \mathbb{Q}^m \to \mathbb{Q}^d$ is surjective and intertwines $N_{\mathcal{C}}$ with M, $\ker P_{\mathcal{C}}$ is $N_{\mathcal{C}}$-invariant and the induced map on $\mathbb{Q}^m/\ker P_{\mathcal{C}}$ is conjugate to M; block triangularization gives $\chi_{N_{\mathcal{C}}} = \chi_{N_{\mathcal{C}}|\ker P_{\mathcal{C}}} \cdot \chi_M$, and Gauss's lemma upgrades divisibility to $\mathbb{Z}[x]$. Finally $\lambda^\top := \ell^\top P_{\mathcal{C}}$ is strictly positive (each $p_j \neq 0$) and $\lambda^\top N_{\mathcal{C}} = \ell^\top MP_{\mathcal{C}} = \beta\lambda^\top$; for an irreducible nonnegative matrix a positive eigenvector belongs to the Perron root, so $\rho(N_{\mathcal{C}}) = \beta$. □

Remark 5.3 (Closed versus closed nonproductive). For a general recurrent noncoincident SCC one has instead $\beta L(s) = \sum_t N_{\mathcal{C}}[s,t]L(t) + \operatorname{escape}(s) + \operatorname{leak}(s)$, where escape collects children outside $\mathcal{C}$ and leak collects coincidence children; then $N_{\mathcal{C}}L_{\mathcal{C}} \leq \beta L_{\mathcal{C}}$ componentwise and $\rho(N_{\mathcal{C}}) \leq \beta$, with equality if and only if escape and leak vanish identically. Closed *productive* SCCs with $\rho(N_{\mathcal{C}}) < \beta$ exist; every statement below that uses $\rho(N_{\mathcal{C}}) = \beta$ requires nonproductivity.

5.3. **Boundary synchronization and locality.** Let $\sigma_+(a)$ and $\sigma_-(a)$ be the first and last letters of $\sigma(a)$. For $h : \mathcal{A} \to \mathcal{A}$, an ordered pair (a, b) is *h-synchronizing* if $h^m(a) = h^m(b)$ for some $m \geq 0$; write $\operatorname{Sync}_\pm$ for the $\sigma_\pm$-synchronizing pairs. For a zero return k of (u, v) with $0 < k < n$, its *right pair* is (u_{k+1}, v_{k+1}) and its *left pair* is (u_k, v_k).

Lemma 5.4 (Boundary synchronization; [*Theorem*]). *If some zero return of $(\sigma^n(u), \sigma^n(v))$, for some $n \geq 0$, has right pair in Sync_+ or left pair in Sync_-, then the state (u, v) is productive.*

Proof. Let K be the zero return and (a, b) its right pair. After m further inflations K is inherited to a zero return K_m (Lemma 2.10) whose right pair is $(\sigma_+^m(a), \sigma_+^m(b))$. If this is (c, c), then $K_m + 1$ is also a zero return and the block between them is (c, c), a coincidence block among the depth-$(n+m)$ descendants of (u, v) (Lemma 2.12). For the left pair, use σ_- and subtract e_c from the equal prefix vectors at K_m. □

Theorem 5.5 (Global endpoint synchronization; [*Theorem*], no PIP hypothesis). *If σ_+ (or σ_-) is globally synchronizing, i.e. some iterate is constant, then every balanced pair is productive and Conjecture 2.15 holds for σ.*

Proof. Position 0 (resp. n) is a zero return of every balanced pair, with right pair (u_1, v_1) (resp. left pair (u_n, v_n)); apply Lemma 5.4. □

Lemma 5.6 (Newborn locality; [*Theorem*]). *Let k be a newborn zero return of $(\sigma(u), \sigma(v))$. There is a unique block T_j of $\operatorname{red}(u, v)$ and an interior zero return r of $\sigma(T_j)$ with $k = B_j + r$, B_j the inherited image of the left end of T_j, and the left and right pairs of k in $\sigma(u, v)$ equal those of r in $\sigma(T_j)$.*

Proof. k lies strictly between consecutive inherited returns $B_j < k < B_{j+1}$; subtracting the prefix equalities at B_j and k shows $r = k - B_j$ is a zero return of $\sigma(T_j)$, and adjacent letters are unchanged by translation. □

Definition 5.7 (Synchronization targets). For a strict component $\mathcal{C}$:

- **C2**: some state $T \in \mathcal{C}$ and some $n \geq 1$ give a zero return of $\sigma^n(T)$ with right pair in Sync_+ or left pair in Sync_-.
- **C3-local**: some $T \in \mathcal{C}$ gives an *interior* zero return of $\sigma(T)$ with right pair in Sync_+ or left pair in Sync_-.

Proposition 5.8 (One-way chain; [*Theorem*])**.** *For every strict component, C3-local ⇒ C2 ⇒ contradiction with nonproductivity. Hence if C3-local holds for every hypothetical strict component of every PIP substitution with finite $\mathcal{B}_\sigma$, then Conjecture 2.15 holds for all of them. No converse is asserted.*

Proof. C3-local is the case $n = 1$ of C2, and C2 contradicts nonproductivity by Lemma 5.4. (Lemma 5.6 shows that a synchronizing newborn zero return at depth n localizes to a one-step interior zero return of a depth-$(n-1)$ block, which lies in $\mathcal{C}$ by closure; so for strict components C2 and C3-local are in fact equivalent, and the inherited case is covered by the boundary pairs of states of $\mathcal{C}$.) □

The program of proving C3-local for every strict component is referred to below as the C4 program. It is open (Open Problem 5.29).

5.4. **Endpoint dynamics on three letters.** For $d = 3$ the 27 self-maps of $\mathcal{A}$ fall into seven conjugacy classes of functional graphs, listed with the recurrent nonsynchronizing ordered pairs of the product map $h \times h$:

Type	Representative	Functional graph	Class size	Nonsync. core
A	$(0,0,0)$	fixed point with two leaves	3	$\emptyset$
B	$(0,0,1)$	fixed point with a tail of length two	6	$\emptyset$
C	$(0,0,2)$	two fixed points, one leaf	6	2 pairs
D	$(0,1,2)$	identity	1	6 pairs
E	$(0,2,1)$	fixed point and a 2-cycle	3	6 pairs
F	$(1,0,0)$	2-cycle with one leaf	6	2 pairs
G	$(1,2,0)$	3-cycle	2	6 pairs

Proposition 5.9 ([*Theorem*])**.** *Every self-map of a three-letter alphabet is conjugate to exactly one of A–G. Types A and B are globally synchronizing. For every h and every ordered pair (a, b), either the pair synchronizes, or it lies in the recurrent core of $h \times h$, or it enters that core after exactly one step; recurrent cores have periods 1, 2 or 3. Let $a \sim_h b$ iff $h^n(a) = h^n(b)$ for some n; h induces a permutation of the quotient $Q_h = \mathcal{A}/{\sim_h}$, and (a, b) synchronizes iff $a \sim_h b$.*

Proof. A functional graph on three vertices is a union of cycles with in-trees; enumerating cycle types (1), $(1, 1)$, $(1, 1, 1)$, $(1, 2)$, (2), (3) and tree attachments gives exactly the seven rows, whose class sizes sum to 27. The remaining assertions are checked on the representatives; both the exhaustive enumeration and the checks were also carried out mechanically. □

Corollary 5.10 (Signature bound; [*Theorem*])**.** *Let $\mathcal{C}$ be a strict component, $d = 3$. For $T = (u, v) \in \mathcal{C}$ the unordered pairs $\{[u_1], [v_1]\} \subset Q_{\sigma_+}$ and $\{[u_n], [v_n]\} \subset Q_{\sigma_-}$ consist of two distinct classes. The number of such* endpoint signatures *is at most $\binom{|Q_+|}{2}\binom{|Q_-|}{2} \le 9$, and every strict component contains a cycle of states whose signatures repeat with period at most 1, 3 or 9 according as the two quotient sizes are $(2, 2)$, mixed, or $(3, 3)$.*

Proof. Equal classes would make position 0 (resp. n) a synchronizing zero return (Lemma 5.4). The signature is invariant under side swap, hence well defined on states. Along an infinite path in the finite signature graph a signature repeats. □

Imported Theorem 5.11 (Barge–Diamond [9]; [*Imported*])**.** *Let σ be a substitution of Pisot type (primitive, irreducible characteristic polynomial, Pisot dilation) on $d \ge 2$ letters. There exist letters $a \ne b$ and $n \ge 1$ such that $\sigma^n(a) = pcs$, $\sigma^n(b) = p'cs'$ with $c \in \mathcal{A}$ and $\pi(p) = \pi(p')$ (the pair $\{a, b\}$ is* eventually coincident*). For $d = 2$ every pair of distinct letters is eventually coincident (strong coincidence).*

Theorem 5.12 (Barge–Diamond eliminator; [*Theorem*], $d = 3$)**.** *Let $\mathcal{C}$ be a closed nonproductive set of states of a PIP substitution on three letters (finiteness not required). Then no zero return of any $\sigma^n(T)$, $T \in \mathcal{C}$, has an eventually coincident right pair; consequently neither σ_+ nor σ_- is of type G, and, fixing one eventually coincident pair $\{a, b\}$ with complementary letter c, every right pair at a zero return of every $\sigma^n(T)$ is $\{a, c\}$ or $\{b, c\}$.*

Proof. If a zero return K has right pair (a, b) eventually coincident at level n, then after n inflations the blocks $\sigma^n(a)$ and $\sigma^n(b)$ start at the inherited zero return, and the position $|p| = |p'|$ after it is a zero return followed by (c, c): a coincidence block, contradicting nonproductivity (Lemma 2.12). Now let $\{a, b\}$ be the pair of Theorem 5.11. The first letters of any $T \in \mathcal{C}$ are distinct and form the right pair of the zero return 0; the first block of $\mathrm{red}(\sigma(T))$ has first letters $(\sigma_+(u_1), \sigma_+(v_1))$ and lies in $\mathcal{C}$. Iterating, every pair $\{\sigma_+^k(u_1), \sigma_+^k(v_1)\}$ is a non-eventually-coincident pair. If σ_+ were a 3-cycle, its action on the three unordered pairs would be transitive and $\{a, b\}$ would occur, a contradiction. The suffix statement follows by applying the same argument to the reversed substitution $a \mapsto \overline{\sigma(a)}$, which has the same incidence matrix and whose balanced-pair automaton is the image of $\mathcal{B}_\sigma$ under word reversal, with σ_- as its prefix map. The last assertion holds because only three unordered pairs exist. □

Together with Theorem 5.5, a strict component on three letters has both endpoint maps of type C, D, E or F. This is a necessary condition; it is not asserted that these types are realizable by strict components.

5.5. **Orientation and signed first-defect intertwiners.** Fix normalized representatives $T = (u_T, v_T)$ of the states of a strict component $\mathcal{C}$. The blocks of $\mathrm{red}(\sigma(u_T), \sigma(v_T))$ are raw pairs $R_1, \ldots, R_m$; each R_i equals (u_U, v_U) or (v_U, u_U) for its state U, and we set $\epsilon(T \to U_i) = +1$ or -1 accordingly. Let A_{ij}, B_{ij} count the occurrences with sign $+1$, -1 of T_i among the children of T_j, so $N_{\mathcal{C}} = A + B$; put $S = A - B$.

Proposition 5.13 (Orientation cocycle; [*Theorem*])**.** *The signs form a $\mathbb{Z}/2$-cocycle on the child graph. A* gauge *$g : \mathcal{C} \to \{\pm 1\}$ with $g(U) = g(T)\epsilon(T \to U)$ on every occurrence exists iff every closed walk has sign product $+1$; then, after reorienting each state by g, the two side words define morphisms $U, V : \mathcal{C}^* \to \mathcal{A}^*$ with $\sigma \circ U = U \circ \tau_{\mathcal{C}}$, $\sigma \circ V = V \circ \tau_{\mathcal{C}}$, $\pi \circ U = \pi \circ V$, and $U(T) \neq V(T)$ for all T. If no gauge exists, the oriented double cover $\mathcal{C} \times \{\pm 1\}$ with lifted children $(T, o) \to (U, o\epsilon)$ is strongly connected and carries a single morphism W with $\sigma \circ W = W \circ \tilde{\tau}_{\mathcal{C}}$ commuting with the deck involution. In all cases $\rho(S) \leq \beta$; if some parent–child pair occurs with both signs then $\rho(S) < \beta$; and if $N_{\mathcal{C}}$ is primitive then $\rho(S) = \beta$ forces S to be diagonally similar to $N_{\mathcal{C}}$ (gauge case) or to $-N_{\mathcal{C}}$ (anti-gauge case).*

Proof. Swapping a parent's sides swaps all its children's sides, so signs compose along paths and the gauge equation is the standard voltage-graph condition; the word identities follow because with a gauge every raw child appears in its chosen orientation. For the spectral statements: $|S| \leq N_{\mathcal{C}}$ entrywise, and for irreducible nonnegative $N_{\mathcal{C}}$ the Wielandt comparison gives $|\lambda| \leq \rho(N_{\mathcal{C}}) = \beta$ (Theorem 5.2) for every eigenvalue λ of S, strictly if $|S| \neq N_{\mathcal{C}}$; in the equality case, if $\lambda = \beta e^{i\varphi}$ is an eigenvalue of S of maximal modulus, Wielandt's theorem [18, Thm. 8.4.5] gives $|S| = N_{\mathcal{C}}$ and $S = e^{i\varphi} D N_{\mathcal{C}} D^{-1}$ for a diagonal unitary $D = \mathrm{diag}(d_i)$, so every edge sign is $e^{i\varphi} d_i d_j^{-1}$ and the sign product around a directed cycle of length L is $e^{i\varphi L} \in \{\pm 1\}$. If $N_{\mathcal{C}}$ is primitive the cycle lengths have greatest common divisor 1, so $e^{2i\varphi} = 1$ and $e^{i\varphi} = \pm 1$; since S and $\pm N_{\mathcal{C}}$ are real with the same support and the graph is strongly connected, the d_i may then be rescaled to lie in $\{\pm 1\}$. □

For a word w and $x = x_1 \cdots x_r \in \mathcal{A}^r$ let $N_x(w)$ be the number of occurrences of x as a scattered subword of w, and let $N_r(w) = (N_x(w))_{x \in \mathcal{A}^r} \in \mathbb{Z}^{d^r}$. For a balanced pair define the *degree-r defect* $K_r(u, v) = N_r(u) - N_r(v)$; $K_1 \equiv 0$ on balanced pairs, and $K_r(v, u) = -K_r(u, v)$.

For $r = 2$ and a balanced pair with $n_a = |u|_a = |v|_a$ one has $K_{aa} = \binom{n_a}{2} - \binom{n_a}{2} = 0$ and $K_{ab} + K_{ba} = n_a n_b - n_a n_b = 0$, so K_2 is antisymmetric; for $d = 3$ we identify it throughout with the vector $(K_{12}, K_{13}, K_{23}) \in \Lambda^2\mathbb{Z}^3$, so that $K_{ji} = -K_{ij}$ (this is the instance $\mathrm{Lie}_2 = \Lambda^2$ of Theorem 5.15 below, and the identification is used in Sections 5.6 and 5.7). For a strict component let $r(\mathcal{C}) = \min\{r : K_r(T) \neq 0 \text{ for some } T \in \mathcal{C}\}$, finite since distinct words of length n have different N_n.

Lemma 5.14 (Concatenation and substitution; [*Theorem*])**.** *(i) $N_r(xy) = \sum_{k=0}^{r} N_k(x) \otimes N_{r-k}(y)$ (with $N_0 = 1$). (ii) $N_r(\sigma(w)) = M^{\otimes r} N_r(w) + \Lambda_r(N_1(w), \ldots, N_{r-1}(w))$ where Λ_r is linear with coefficients depending only on σ. (iii) The functional $x \mapsto N_x(w)$ is a character of the shuffle algebra (Ree's theorem* [33, Thm. 3.6]*): $N_{x ⧢ y}(w) = N_x(w)N_y(w)$.*

Proof. (i) An occurrence splits into its parts in x and in y. (ii) Group occurrences in $\sigma(w)$ by the multiset of source positions used; the occurrences using r distinct source positions contribute $\sum_y \prod_j M_{x_j y_j} N_y(w) = (M^{\otimes r} N_r(w))_x$, and those using fewer source positions are linear in lower-degree counts of w with coefficients given by scattered-subword counts inside the images $\sigma(a)$. (iii) is Ree's theorem. ☐

Theorem 5.15 (First-defect intertwiner; [*Theorem*]; part of Theorem 3.4)**.** *Let $r = r(\mathcal{C})$ and let Q_r have the vectors $K_r(T)$, $T \in \mathcal{C}$, as columns. Then every $K_r(T)$ lies in the free Lie component $\mathrm{Lie}_r(\mathbb{Q}^d) \subset (\mathbb{Q}^d)^{\otimes r}$, and $Q_r S = \Phi_r Q_r$ with $\Phi_r = M^{\otimes r}|_{\mathrm{Lie}_r}$. In particular $\operatorname{im} Q_r$ is a nonzero rational Φ_r-invariant subspace on which the induced action is a quotient of S, so $\rho(\Phi_r|_{\operatorname{im} Q_r}) \leq \beta$.*

Proof. For $T \in \mathcal{C}$ all $K_k(T)$, $k < r$, vanish, so Lemma 5.14(ii) gives $K_r(\sigma(T)) = M^{\otimes r} K_r(T)$, and (i) applied to $\sigma(u_T) = R_1 \cdots R_m$ (top sides) and to the bottom sides, with all lower-degree cross terms equal on both sides, gives $K_r(\sigma(T)) = \sum_i \epsilon_i K_r(U_i)$, which is $Q_r S$ column by column. By (iii), $K_r(T)$ annihilates every proper shuffle $x ⧢ y$ with $|x|, |y| \geq 1$, $|x| + |y| = r$; the annihilator in degree r of the proper shuffles is the space of primitive elements of the concatenation Hopf algebra, which is Lie_r [33, Thm. 3.1]. Since $M^{\otimes r}$ is an algebra automorphism preserving Lie_r, the identity restricts. ☐

Theorem 5.16 (Degrees two and three, $d = 3$; [*Theorem*])**.** *Let $d = 3$ and $\mathcal{C}$ strict.*

(i) $\chi_{\Lambda^2 M}(x) = x^3 - Ux^2 + \det(M)\operatorname{tr}(M)x - \det(M)^2$*, where* $\chi_M = x^3 - \operatorname{tr}(M)x^2 + Ux - \det M$*, is irreducible over* $\mathbb{Q}$*, and* $\rho(\Lambda^2 M) = \beta|\beta_2| < \beta$*.*

(ii) If $r(\mathcal{C}) = 2$ *then* $\operatorname{rank}_{\mathbb{Q}} Q_2 = 3$*,* $\chi_{\Lambda^2 M} \mid \chi_S$*, and* $|\mathcal{C}| \geq \deg \operatorname{lcm}(\overline{\chi_M}, \overline{\chi_{\Lambda^2 M}}) \in \{3, 4, 5, 6\}$*, the reductions being taken modulo* 2*; the value is* 6 *whenever* $\det M$ *is odd and* $\operatorname{tr} M \not\equiv U \pmod 2$*.*

(iii) If $r(\mathcal{C}) = 3$ *then* $\operatorname{im} Q_3 \subseteq \ker(\Phi_3 - \det M)$*, a two-dimensional space, so* $\operatorname{rank} Q_3 \leq 2$*.*

Proof. (i) The roots of $\chi_{\Lambda^2 M}$ are $\beta_i\beta_j = \det M/\beta_k$; a rational one would make some β_k rational. The moduli are $\beta|\beta_2| \geq \beta|\beta_3|$ and $|\beta_2\beta_3| < 1 < \beta|\beta_2|$ (as $|\det M| = \beta|\beta_2\beta_3| \geq 1$), and $\beta|\beta_2| < \beta$. (ii) $\mathrm{Lie}_2 = \Lambda^2$, so $\operatorname{im} Q_2$ is a nonzero $\Lambda^2 M$-invariant rational subspace of $\Lambda^2\mathbb{Q}^3$, hence everything by irreducibility; then $\Lambda^2 M$ is a quotient of S and $\chi_{\Lambda^2 M} \mid \chi_S$ in $\mathbb{Z}[x]$ (Gauss). Since $S \equiv N_{\mathcal{C}} \pmod 2$, $\chi_S \equiv \chi_{N_{\mathcal{C}}} \pmod 2$ is divisible in $\mathbb{F}_2[x]$ by both reductions; the table of the eight parity classes is a direct computation in $\mathbb{F}_2[x]$, and when $\det M$ is odd the difference of the two reductions is $x(x+1)$ while neither vanishes at 0 or 1. (iii) For $d = 3$, $\mathrm{Lie}_3(V) \cong S_{(2,1)}(V)$ has dimension 8 and Φ_3 has eigenvalues $\beta_i^2\beta_j$ $(i \neq j)$ and $\det M$ with multiplicity two; Φ_3 is semisimple since χ_M is. The Galois-stable rational constituents of the non-scalar part are one sextic factor or two cubic factors, each containing a weight of modulus $\beta^2|\beta_2|$ or $\beta^2|\beta_3|$; both exceed β because $\beta|\beta_2| \geq \sqrt{|\det M|} \geq 1$ and $\beta|\beta_3| = |\det M|/|\beta_2| > 1$. A rational invariant subspace of spectral radius $\leq \beta$ therefore has zero component in every non-scalar constituent, hence lies in the $\det M$-eigenspace; apply Theorem 5.15. ☐

Theorem 5.17 (Higher degrees, $d = 3$; [*Theorem*]). *Let $d = 3$, $\mathcal{C}$ strict, $r = r(\mathcal{C}) \geq 4$. Every rational irreducible Φ_r-constituent of $\mathrm{Lie}_r(\mathbb{Q}^3)$ is a Galois orbit of weights $\beta_1^a\beta_2^b\beta_3^c$ with $a+b+c = r$, and its spectral radius is at most β only if, up to permutation, (a, b, c) is one of: (m, m, m) if $r = 3m$; $(m + 1, m, m)$, or $(m + 1, m + 1, m - 1)$ in the case of complex conjugate β_2, β_3 with $|\det M| = 1$, if $r = 3m + 1$; $(m + 1, m + 1, m)$ if $r = 3m + 2$. In particular, if $r = 4$ then $|\det M| = 1$ and $\rho(S) = \beta$. If moreover $|\mathcal{C}| = 3$: in the Perron-extremal case $\rho(S) = \beta$ one has $r \equiv 1 \pmod 3$, $|\det M| = 1$, and $\operatorname{im} Q_r$ is the constituent $\det(M)^m \otimes M$; in the case $\rho(S) < \beta$ one has $r \equiv 0$ or $2 \pmod 3$.*

Proof sketch. $\mathrm{Lie}_r(\mathbb{Q}^3)$ is a polynomial functor of $V = \mathbb{Q}^3$, so its weights under a semisimple M with eigenvalues β_i are the monomials $\prod \beta_i^{a_i}$ with multiplicities given by the generalized Witt formula $m(a, b, c) = \frac{1}{r}\sum_{t|\gcd(a,b,c)} \mu(t)\frac{(r/t)!}{(a/t)!(b/t)!(c/t)!}$ [33, Cor. 4.15]. A rational invariant subspace is a sum of Galois orbits of weight spaces, and every orbit of an exponent vector contains an assignment with its largest exponent a on β. With $x = \log\beta$, $y = -\log|\beta_2|$, $z = -\log|\beta_3|$, $\delta = \log|\det M| = x - y - z \geq 0$, the logarithm of that weight's modulus relative to β is $(a - 1 - b)y + (a - 1 - c)z + (a - 1)\delta$ (or with y, z exchanged), which is strictly positive unless the exponent vector has the listed shapes; the case analysis is elementary. For $r = 4$ the shapes are $(2, 1, 1)$ (operator $\det(M) \cdot M$, spectral radius $|\det M|\beta$) and, only for complex stable roots with $|\det M| = 1$, $(2, 2, 0)$; both have spectral radius $\geq \beta$, forcing $\rho(S) = \beta$ and $|\det M| = 1$ through Theorem 5.15. If $|\mathcal{C}| = 3$ then $N_{\mathcal{C}} = P_{\mathcal{C}}^{-1}MP_{\mathcal{C}}$ (Theorem 5.2) is primitive, so in the extremal case $\rho(S) = \beta$ Proposition 5.13 makes S diagonally similar to $\pm N_{\mathcal{C}}$, with the irreducible cubic characteristic polynomial of $\pm M$; any nonzero Q_r is then injective and its image is a three-dimensional constituent similar to $\pm M$, and comparing determinants and moduli excludes every shape except $(m + 1, m, m)$ with $|\det M| = 1$. In the non-extremal case $\rho(S) < \beta$, the shapes $(m+1, m, m)$ and $(m+1, m+1, m-1)$, whose spectral radius is at least β, are excluded, leaving $r \equiv 0, 2 \pmod 3$. We do not know whether a three-state strict component can have $\rho(S) < \beta$ with $r \in \{2, 3\}$; Theorems 5.16(ii),(iii) constrain but do not exclude these cases. □

Remark 5.18. Degree five is the first degree at which a genuinely sub-Perron higher constituent survives in the unimodular case (shape $(2, 2, 1)$, operator $\det(M)\Lambda^2 M$, spectral radius $\beta|\beta_2|$). The sieves are necessary conditions on the algebra of a strict component; none of them, alone or jointly, has been shown inconsistent for every possible component. Their role is to shrink the hypothetical object and to calibrate the finite censuses of Section 6.

5.6. **Ordered factorization: the mid-area identity.** For $w \in \mathcal{A}^*$ with $d = 3$ let $\mathrm{area}(w) = (N_{12} - N_{21}, N_{13} - N_{31}, N_{23} - N_{32})(w) \in \Lambda^2\mathbb{Z}^3$ and let C_σ have columns $\mathrm{area}(\sigma(a))$.

Proposition 5.19 ([*Theorem*]). *(i)* $\mathrm{area}(\sigma(w)) = (\Lambda^2 M)\,\mathrm{area}(w) + C_\sigma\pi(w)$. *(ii)* $\mathrm{area}(x_1 \cdots x_m) = \sum_r \mathrm{area}(x_r) + \sum_{r<s}\pi(x_r) \wedge \pi(x_s)$. *(iii) For a balanced pair $T = (u, v)$ put $H(T) = \mathrm{area}(u) + \mathrm{area}(v)$; then, with the identification of K_2 with $(K_{12}, K_{13}, K_{23}) \in \Lambda^2\mathbb{Z}^3$ of Section 5.5,* $\mathrm{area}(u) - \mathrm{area}(v) = 2K_2(T)$, $K_2(\sigma(T)) = (\Lambda^2 M)K_2(T)$, *and if* $\mathrm{red}(\sigma(T)) = (R_1, \ldots, R_m)$ *then*

$$(\Lambda^2 M)H(T) + 2C_\sigma\pi(T) = \sum_r H(R_r) + 2\sum_{r<s}\pi(R_r) \wedge \pi(R_s).$$

Proof. (i) and (ii) count degree-two occurrences by whether both positions lie in one image, resp. one block. (iii) The cross terms in (ii) are the same on both sides since the blocks are balanced; subtracting gives block additivity of K_2 and, with (i), the push-forward; adding gives the displayed identity. □

The identity retains the *order* of the children while forgetting their orientation. It is a necessary condition for a proposed derived substitution to be the actual factorization, and it is the first invariant in this paper that rejects the synthetic template of Section 7.1. For $|\mathcal{C}| = 3$ the Sylvester operator $X \mapsto (\Lambda^2 M)X - XN_{\mathcal{C}}$ is invertible over $\mathbb{Q}$ (Theorems 5.2 and 5.16(i) give $\operatorname{spec} N_{\mathcal{C}} \cap \operatorname{spec} \Lambda^2 M = \emptyset$, because $\beta_i = \beta_j\beta_k$ would force $\beta_i^2 = \det M \in \mathbb{Q}$), so a prescribed ordered child system forces a unique rational H, which must be integral with $H \pm 2Q_2$ satisfying the elementary parity and magnitude bounds $|\operatorname{area}_{ij}(w)| \le n_i n_j$, $\operatorname{area}_{ij}(w) \equiv n_i n_j \pmod 2$. We regard the three-state calculations as calibration only.

5.7. **The wedge dichotomy and the spectral route.** Let $d = 3$. By Theorem 5.16(i), $\Lambda^2 M$ is semisimple with a dominant eigenvalue of modulus $\beta|\beta_2|$ (simple, or a complex pair); let Π_{dom} be the corresponding real spectral projection.

Proposition 5.20 (Wedge dichotomy; [*Theorem*]; Proposition 3.6)**.** *Let $\mathcal{C}$ be a closed non-productive SCC (finite or not), $V_{\mathcal{C}} = \operatorname{span}_{\mathbb{Q}}\{K_2(T) : T \in \mathcal{C}\}$. Then $(\Lambda^2 M)V_{\mathcal{C}} = V_{\mathcal{C}}$, hence $V_{\mathcal{C}} \in \{0, \Lambda^2\mathbb{Q}^3\}$. If $V_{\mathcal{C}} \neq 0$ then for every nonzero linear functional φ on $\Lambda^2\mathbb{R}^3$ some $T \in \mathcal{C}$ has $\varphi(K_2(T)) \neq 0$; in particular $\Pi_{\mathrm{dom}} K_2(T) \neq 0$ for some $T \in \mathcal{C}$, and the three eigencoordinates of $K_2(T)$, $T \in \mathcal{C}$, are not all zero for any eigenvalue.*

Proof. By Proposition 5.19(iii), $(\Lambda^2 M)K_2(T) = K_2(\sigma(T)) = \sum_i \epsilon_i K_2(U_i)$ with $U_i \in \mathcal{C}$ (closure and nonproductivity), so $(\Lambda^2 M)V_{\mathcal{C}} \subseteq V_{\mathcal{C}}$, with equality since $\Lambda^2 M$ is invertible. Irreducibility of $\chi_{\Lambda^2 M}$ leaves only 0 and $\Lambda^2\mathbb{Q}^3$. The rest is linear algebra. □

Proposition 5.21 (Wedge growth in a finite closed component; [*Conditional*] on finiteness of $\mathcal{C}$)**.** *Let $\mathcal{C}$ be a finite closed recurrent noncoincident SCC (productive or not) and suppose $\Pi_{\mathrm{dom}} K_2(s) \neq 0$ for some $s \in \mathcal{C}$. Then $\rho(N_{\mathcal{C}}) \ge \beta|\beta_2|$.*

Proof. By Proposition 5.19(iii) and Lemma 2.12, $(\Lambda^2 M)^k K_2(s) = \sum_{c \in D_k} \pm K_2(c)$ over the depth-k descendants of s; coincidence descendants contribute 0, and by closure every noncoincident descendant lies in $\mathcal{C}$. Since Π_{dom} commutes with $\Lambda^2 M$ and the restriction of $\Lambda^2 M$ to the dominant real eigenspace is $\beta|\beta_2|$ times an isometry in a suitable norm, $\|(\Lambda^2 M)^k K_2(s)\| \ge c(\beta|\beta_2|)^k$. The left side is at most $\max_{T\in\mathcal{C}} \|K_2(T)\|$ times the number $\|N_{\mathcal{C}}^k e_s\|_1$ of noncoincident depth-k descendants, which is $O(k^m \rho(N_{\mathcal{C}})^k)$. □

The spectral route to Conjecture 2.15 pursued in this program was: (a) a closed nonproductive carrier $\mathcal{C}$ has some $\Pi_{\mathrm{dom}} K_2 \neq 0$ ("concentration"); (b) Galois conjugacy then makes all wedge eigenfunctionals nonzero on $\mathcal{C}$ ("span-rich"); (c) a span-rich closed carrier is productive. Proposition 5.20 shows that (a) is *equivalent* to the statement that a closed nonproductive carrier cannot have $K_2 \equiv 0$, that (b) follows from (a) with no Galois argument, and that (c) is equivalent to the statement that a closed nonproductive carrier cannot have $K_2 \not\equiv 0$. Proposition 5.21 gives $\rho(N_{\mathcal{C}}) \ge \beta|\beta_2|$, which is consistent with $\rho(N_{\mathcal{C}}) = \beta$ (Theorem 5.2) and yields no contradiction. We have found no proof of (a) or of (c), and we formulate both as what they are:

Open Problem 5.22 (Concentration; [*Open*])**.** *Show that a strict component $\mathcal{C}$ on three letters cannot have $K_2 \equiv 0$; equivalently, that every strict component has first defect degree two. Theorems 5.16(iii) and 5.17 constrain the cases $r(\mathcal{C}) \ge 3$ but do not exclude them.*

Open Problem 5.23 (Wedge productivity; [*Open*])**.** *Show that a strict component $\mathcal{C}$ on three letters cannot have $K_2 \not\equiv 0$; equivalently, that every strict component has first defect degree at least three.*

Together, Open Problems 5.22 and 5.23 are exactly the nonexistence of strict components on three letters, split by first defect degree; the spectral route therefore contains no proved implication from wedge data to productivity, and we do not regard either problem as easier than the other. The exhaustive census of Section 6 finds that first defect degree two is the only case realized by reachable states except for 24 states of degree three, so Open Problem 5.23 is the case that the finite evidence points to as the main one; this is an empirical remark, not a reduction.

5.8. **Uniform return and recognizability machinery.** Let $\mathcal{C}$ be a finite set of noncoincident irreducible states closed under children (a strict component qualifies) and $B_{\mathcal{C}} = \max_{T\in\mathcal{C}} |T|$.

Lemma 5.24 (Uniform return gaps; [*Theorem*], no Pisot hypothesis)**.** *For every $T \in \mathcal{C}$ and $n \geq 0$, consecutive zero returns of $\sigma^n(T)$ are at distance at most $B_{\mathcal{C}}$.*

Proof. By Lemma 2.12 the zero returns of $\sigma^n(T)$ include the boundaries of the depth-n descendant blocks, all of which lie in $\mathcal{C}$ and have length $\leq B_{\mathcal{C}}$. □

Theorem 5.25 (Legal ancestry towers; [*Theorem*])**.** *Let σ be PIP, $\mathcal{C}$ as above, $R \geq 0$ and $q \geq 0$. For every $T \in \mathcal{C}$ and all sufficiently large n there is a zero return t of $\sigma^n(T)$ such that t and each of its first q desubstituted source cuts have, on both sides, radius-R contexts contained in the image of a single letter of the base word under the appropriate power of σ; in particular these contexts are factors of the language of σ and Mossé's theorem applies at each of the $q+1$ levels.*

Proof. Put $\Lambda = \max_a |\sigma(a)|$ and $m_0 = R$, $m_{j+1} = (m_j + 1)\Lambda$. A cut at distance $\geq m_{j+1}$ from the ends of an image $\sigma^n(a)$ has source cut at distance $\geq m_j$ from the ends of $\sigma^{n-1}(a)$. The base words u, v have $|u| + |v| + 2$ letter boundaries, each of which forbids at most $2m_q + 1$ output positions; this number is independent of n, whereas by Lemma 5.24 the number of zero returns of $\sigma^n(T)$ is at least $|\sigma^n(u)|/B_{\mathcal{C}}$, which tends to infinity. Any factor of $\sigma^k(a)$ is in the language by primitivity. □

Theorem 5.26 (Derived recognizability; [*Theorem*])**.** *Let $\mathcal{C}$ be a strict component and h the period of the irreducible matrix $N_{\mathcal{C}}$. The cyclic classes $\mathcal{C}_0, \dots, \mathcal{C}_{h-1}$ are preserved by $\tau_{\mathcal{C}}^h$, each restriction $\rho_j = \tau_{\mathcal{C}}^h|_{\mathcal{C}_j}$ is a primitive substitution with Perron eigenvalue β^h, its subshift is aperiodic, and Mossé's theorem applies to each ρ_j: there is a radius $R_{\mathcal{C}}$ such that $\tau_{\mathcal{C}}^h$-supertile boundaries in points of the derived subshifts are determined by radius-$R_{\mathcal{C}}$ derived context.*

Proof. The cyclic decomposition of an irreducible matrix and the Perron root β^h of the diagonal blocks of $N_{\mathcal{C}}^h$ are standard [12]. If some ρ_j had a periodic subshift, the argument of Lemma 2.4 would give $\beta^{hq} \in \mathbb{Z}$ for some q, and the conjugate β_2 would satisfy $|\beta_2|^{hq} \geq 1$, a contradiction. Apply Theorem 2.6. □

Remark 5.27. The Pisot hypothesis is essential: $A \mapsto B$, $B \mapsto AA$ has period 2, Perron root $\sqrt{2}$, and its square restricted to a cyclic class is $A \mapsto AA$, which is periodic; the non-Pisot strict component of Example 7.2 has derived substitution $A \mapsto AB$, $B \mapsto AB$, primitive with periodic subshift.

Proposition 5.28 (Finite relative hierarchy-offset state; [*Theorem*])**.** *Let $\mathcal{C}$ be a strict component and fix a context radius. For a zero return of $\sigma^n(T)$ desubstituted to source depth k, record: the source defect x_k; the outgoing correction digit; the source-index difference $i - j = \langle \mathbf{1}, x_k\rangle$; the difference of the indices of the derived blocks of $\tau_{\mathcal{C}}^k(T)$ containing the two cuts; the in-block offsets; the two block states with orientations; and the radius-R derived context. This datum ranges over a finite set independent of n and k, and the block-index difference is bounded by $|\langle \mathbf{1}, x_k\rangle|$.*

Proof. Theorem 4.13 bounds x_k; each derived block has positive physical length, so crossing a block boundary costs at least one symbol; all other coordinates are visibly finite. □

Theorems 5.25–5.26 and Proposition 5.28 make both hierarchies (that of σ and that of $\tau_{\mathcal{C}}$) locally recognizable at a recurrent finite-state cut. They do not force the two hierarchies to align, and the negative controls of Section 7 show that neither finiteness nor recurrence of the offset state contradicts strictness by itself.

Open Problem 5.29 (C4: uniform alignment; [*Open*])**.** *Prove that a strict component of a PIP substitution cannot carry a recurrent relative hierarchy-offset state along arbitrarily long legal* *ancestry towers with every newborn boundary endpoint-nonsynchronizing; equivalently, prove C3-local (Definition 5.7) for every strict component, uniformly in $|\mathcal{C}|$.*

5.9. **Formal recurrence versus realization.** Under Hypothesis 2.14 the coincidence rank of σ [10, 6] is greater than one if and only if some fibre of the maximal equicontinuous factor contains two tilings that never coincide, and a recurrent producer-free component of $\mathcal{B}_\sigma$ that is *globally realized* by such a pair would witness this. The converse identification of nontrivial coincidence rank with a globally realized producer-free component has been asserted in working notes of this program; we have not reconstructed a proof and record it as a [*Conjectural bridge*]. The distinction matters in both directions: seeds and states of $\mathcal{B}_\sigma$ need not be legal (Remark 2.13), and a formal recurrent cycle in $\mathcal{B}_\sigma$ is not a pair of tilings. Finite experiments that extend formal producer-free cycles by legal collars and observe that all of them die at a small radius have been carried out in this program; since no theorem bounds the collar radius at which every unrealizable cycle must die, such experiments are not reported here as propositions.

6. Computational propositions

All computations below are exact (integer and rational arithmetic, Sturm sequences for real-root isolation), and each is stated with its search domain and its certificate meaning. None proves a statement about any substitution outside its domain, and none proves G1, Conjecture 2.15, or the Pisot substitution conjecture for any infinite family.

Convention 6.1. The *corpus* $\mathcal{S}$ is the set of substitutions σ on $\mathcal{A} = \{0, 1, 2\}$ with $1 \le |\sigma(a)| \le 3$ for each a (there are $39^3 = 59{,}319$ such substitutions) that satisfy (H1)–(H3), decided exactly: primitivity by positivity of a power of M, irreducibility of the monic cubic by the absence of rational roots, and the Pisot property by exact real-root isolation and, in the complex-conjugate case, by $|\beta_2\beta_3| = |\det M|/\beta < 1$. For each $\sigma \in \mathcal{S}$ the automaton $\mathcal{B}_\sigma$ was constructed by breadth-first closure from the three seeds with a cap of 20,000 states; a run reaching the cap is inconclusive.

Computational Proposition 6.2 (Automaton census; [*Finite computation*])**.** $|\mathcal{S}| = 4{,}554$*, of which* 2,628 *are unimodular and* 1,926 *are not; among the unimodular ones* 648 *have three real roots and* 1,980 *a complex pair; no* χ_M *has zero discriminant. For every* $\sigma \in \mathcal{S}$ *the construction of* $\mathcal{B}_\sigma$ *terminated below the cap. Over* $\mathcal{S}$ *there are* 5,022 *recurrent noncoincident SCCs containing* 369,486 *states; each* σ *has exactly one noncoincident SCC with no noncoincident exit (a* noncoincident sink*), and every one of the* 4,554 *sinks has a coincidence block among its children, hence is productive. Every sink also has a state with a one-step newborn synchronizing zero return, and* 4,278 *sinks have an inherited synchronizing boundary. Exactly* 60 *of the* 5,022 *recurrent noncoincident SCCs have neither a newborn nor an inherited synchronizing boundary among their states; all* 60 *are singletons, none has a coincidence child, and every one has a noncoincident exit. There is no closed nonproductive SCC in the corpus.*

Meaning. For each $\sigma \in \mathcal{S}$ the run certifies that $\mathcal{B}_\sigma$ is finite and that every reachable state is productive; under Imported Theorem 2.16 this is a proof of pure discrete spectrum for that particular σ (the seed-family question of Section 4.8 does not affect this direction). *Limitation.* It says nothing about substitutions with longer images or larger alphabets, and finiteness of 4,554 particular automata is not evidence of a uniform mechanism.

Computational Proposition 6.3 (First-defect census; [*Finite computation*])**.** *Over* $\mathcal{S}$ *the* 385,926 *reachable noncoincident states have first nonzero defect degree* 2 *in* 385,902 *cases and* 3 *in* 24 *cases; no reachable state has* $K_2 = K_3 = K_4 = 0$*. The* 24 *degree-three states occur in* 12 *substitutions, two per substitution, all in the noncoincident sink, with lengths* 9 *and* 19*; up to relabelling and reversal the substitutions are* $0 \mapsto 011,\ 1 \mapsto 2,\ 2 \mapsto 120$ *and its reversal, all with*

$\chi_M = x^3 - 2x^2 - 1$ *and* $\det M = 1$*; for each of the* 24 *states the matrix* $\Theta(K_3) \in \mathrm{Mat}_3(\mathbb{Z})$ *representing* K_3 *in* $\mathrm{Lie}_3 \cong \mathfrak{sl}_3 \otimes \det$ *has* $|\det \Theta| = 2$*,* $\operatorname{tr} \Theta^2 = 6$*, and does not commute with* M*; every one of the* 24 *states reaches a coincidence block within two inflations.*

Meaning. Degree two is the only first-defect degree realized in the corpus apart from 24 states, none of which satisfies the necessary condition $\operatorname{im} Q_3 \subseteq \ker(\Phi_3 - \det M)$ of Theorem 5.16(iii) for membership in a strict component. *Limitation.* First defect degree four does occur for irreducible balanced pairs outside the reachable sets: the pair (122331211223, 231121223312) over $\{1, 2, 3\}$ has $K_1 = K_2 = K_3 = 0$ and $K_4 \neq 0$ (directly checkable; we make no claim that this length is minimal). The census bounds nothing outside $\mathcal{S}$.

Computational Proposition 6.4 (Three-state templates; [*Finite computation*])**.** *Among* $\sigma \in \mathcal{S}$*,* 1,716 *have* $\overline{\chi_M} = \overline{\chi_{\Lambda^2 M}}$ *in* $\mathbb{F}_2[x]$ *(the parity condition of Theorem 5.16(ii) compatible with* $|\mathcal{C}| = 3$*);* 570 *of these have both endpoint maps nonsynchronizing;* 1,554 *of the* 1,716 *satisfy* $\operatorname{tr}(M^k) \geq \operatorname{tr}((\Lambda^2 M)^k)$ *for all* $k \leq 12$ *(a necessary condition for a three-state strict component with* $r(\mathcal{C}) = 2$*, since then* $N_{\mathcal{C}} \sim M$*,* $S \sim \Lambda^2 M$ *and* $\operatorname{tr} N_{\mathcal{C}}^k - \operatorname{tr} S^k \geq 0$*);* 546 *satisfy all three conditions. For none of these* 546 *does* $\mathcal{B}_\sigma$ *contain a recurrent noncoincident SCC of size* 3*; the minimum size of a recurrent noncoincident SCC among them is* 4*.*

Meaning. Arithmetic and endpoint conditions admit hundreds of hypothetical three-state templates that the actual zero-return factorization never realizes. *Limitation.* This is not a theorem excluding three-state strict components; the gap between the template count and the realized count is the object of Open Problems 5.23 and 5.29.

6.1. **Finite computations for Level 2.** The following concern the single substitution τ of Example 4.8 and the single seed $(01, 10)$; nothing is claimed for other substitutions or seeds.

Computational Proposition 6.5 (Renewal census for τ; [*Finite computation*])**.** *Over depths* $1 \leq d \leq 7$ *the pairs* $(\tau^d(01), \tau^d(10))$ *have in total* 601 *interior zero returns, each verified against the level-scaled certificate* $M^d\delta + c = 0$ *of Section 4.4 and against the level-by-level affine recurrence* $x_{t+1} = Mx_t + q_t$*. Let* $\Pi_{r,w}$ *be the finite projection of Corollary 4.10 with source radius* $r = 2$ *and digit window* $w = 1$*. Among the* 601 *cuts,* $\Pi_{2,1}$ *has* 12,466 *colliding unordered pairs. Joining two cuts when their full addresses differ by the insertion of one certified one-level synchronous digit pair on both sides identifies* 946 *of these pairs; refining the remaining* 11,520 *pairs by the classes of the two sidewise prefix translations* $p_{\mathrm{top}}, p_{\mathrm{bot}} \in \mathbb{Z}^3$ *in* $\mathbb{Z}^3/M^k\mathbb{Z}^3$ *leaves* 11,520*,* 10,828*,* 6,897*,* 3,538*,* 1,540*,* 500*,* 83*,* 56*,* 26*,* 1 *unseparated pairs for* $k = 1, \ldots, 10$*. The single surviving pair (the cut at position* 14 *at depth* 7 *and the cut at position* 14 *at depth* 5*) has* equal *sidewise translations* $p_{\mathrm{top}} = p_{\mathrm{bot}} = (5, 6, 3)$*, so no function of these translations separates it; its depth-*7 *affine trace repeats the paired affine state at levels* 1 *and* 3*, and deleting the two edges of that loop yields, digit for digit, the certified depth-*5 *address.*

Meaning. On this finite family every collision of the natural finite invariants is either separated by a residue coordinate of the 2-adic factor or explained by a loop in the affine trace. *Limitation.* A plateau or a separation at a finite k is a property of this family; the pump explanation is a finite certificate of address equality, not of preservation of realizability, and Open Problem 4.15 is untouched.

Computational Proposition 6.6 (Seed-patch overlap graph for τ; [*Finite computation*])**.** *Give the letters of* τ *tile lengths in* $\mathbb{Z}[\beta]$ *proportional to a left Perron eigenvector, place the patches* 01 *and* 10 *with a common left endpoint, and iterate the inflation* $t \mapsto \beta t + q - p$ *on oriented overlap states* (i, j, t) *(top tile* $[0, \ell_i]$*, bottom tile* $[t, t + \ell_j]$*,* p, q *the prefix translations of the selected children), keeping only states with nonempty interior overlap, decided by exact sign determination in* $\mathbb{Q}(\beta)$*. From* 9 *initial overlaps,* 628 *states are reachable, every one of which reaches a coincidence* $(i, i, 0)$*.*

Meaning. The seed-patch overlap graph of τ is finite and coincident, in agreement with the balanced-pair verdict for τ. *Limitation.* This graph is generated from two seed patches, not from all realized overlaps of two tilings in a fibre; its agreement with the balanced-pair automaton on one specimen is evidence for the dictionary of Section 4.7, not a proof of it, and it bounds no chain length.

Empirical Observation 6.7 ([*Empirical*]). Across $\mathcal{S}$: no run reached the cap; every noncoincident sink is productive by a direct coincidence child; the first defect degree is two on all but 24 reachable states; and for τ the finite projections above separate all but a regular family. These patterns motivated the formulations of Open Problems 4.5, 4.15 and 5.23. They are not propositions.

7. Obstructions, counterexamples, and retired routes

Example 7.1 (Cycle exclusion is false; [*Finite computation*])**.** For the flipped Tribonacci substitution $1 \mapsto 21$, $2 \mapsto 31$, $3 \mapsto 1$ (PIP, $\chi_M = x^3 - x^2 - x - 1$), $\mathcal{B}_\sigma$ contains the closed recurrent noncoincident SCC $\{(12,21),(13,31),(213,312),(1213,3121)\}$ with the directed cycle $(12,21) \to (213,312) \to (1213,3121) \to (12,21)$ of period 3 in which every child taken is the leftmost one, so the cycle displacement vanishes (the state $(13,31)$ has the single noncoincident child $(12,21)$). The component is productive: each inflation of each of its states produces the coincidence block $(1,1)$. Hence "no recurrent noncoincident cycle" and "noncoincident cycles have nonzero displacement" are false as general statements, and productivity, not exclusion, is the correct target.

Example 7.2 (Non-Pisot strict component; [*Finite computation*])**.** For $1 \mapsto 2$, $2 \mapsto 123$, $3 \mapsto 2$ (primitive, singular incidence matrix, not Pisot: $\chi_M = x(x-2)(x+1)$, Perron root 2) the states $A = (12,21)$, $B = (23,32)$ form a closed nonproductive recurrent SCC with signed children $A \to A^-B^+$, $B \to A^+B^-$, so $N_{\mathcal{C}} = \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, $S = \left(\begin{smallmatrix}-1&1\\1&-1\end{smallmatrix}\right)$; the cocycle is nontrivial but anti-gauge, and $\rho(S) = 2 = \rho(N_{\mathcal{C}})$. Zero returns of $\sigma^n(A)$ have gap exactly 2 at every depth, the derived substitution is $A \mapsto AB$, $B \mapsto AB$, and the relative offset state is periodic. Hence bounded return gaps, a finite offset state, legal towers, and a nontrivial cocycle are each compatible with strictness in the absence of the Pisot hypothesis, and any proof of Open Problem 5.29 must use it.

Example 7.3 (Nonzero ancestry loop in a PIP substitution; [*Finite computation*])**.** For Tribonacci $1 \mapsto 12$, $2 \mapsto 13$, $3 \mapsto 1$ and the seed $(12,21)$, position 51 at depth 6 is a zero return whose source-defect ancestry is $(1,-1,0),(0,1,-1),(-1,0,1),(1,-1,0),(0,1,-1),(0,0,1),(0,0,0)$, with a repeated nonzero defect; its radius-one decorated germ repeats exactly at levels 1 and 4. Thus Theorem 4.13 cannot be strengthened to "ancestry cycles are trivial", even for productive PIP substitutions.

Example 7.4 (Retired endpoint cone; [*Finite computation*])**.** The four irreducible balanced pairs over $\{0,1,2\}$ with $u_1 = 0, v_1 = 1, u_9 = 1, v_9 = 0$,

$$(021020001, 100002120),\ (010112021, 120021110),\ (011100021, 120001110),\ (022211001, 100112220),$$

have K_2 vectors $q_1 = (2,-5,-2)$, $q_2 = (2,3,4)$, $q_3 = (-2,4,2)$, $q_4 = (1,-3,-9)$ with $50q_1 + 8q_2 + 61q_3 + 6q_4 = 0$. Hence no linear functional is positive on the K_2 of all irreducible pairs with a prescribed outer endpoint class, and the "positive cone" approach to degree two is closed.

7.1. Synthetic template.

There is an explicit ordered three-state child system for the PIP substitution $1 \mapsto 2$, $2 \mapsto 3$, $3 \mapsto 132$ (with actual irreducible balanced states as vertices and endpoint types G and F) that satisfies the Parikh intertwiner with rank $P = 3$, the signed degree-two intertwiner with rank $Q_2 = 3$, nonnegative orientation counts, and the endpoint quotient constraints, yet is not the actual zero-return factorization: its mid-area residuals (Proposition 5.19(iii)) are $(0,0,-4)$, $(0,-4,-4)$, $(-4,-8,-12)$, and the unique rational solution H of the Sylvester system is half-integral. The actual factorizations of the same states have zero residual. This template is the reason the order-sensitive identity is retained and the reason that no invariant coarser than the factorization itself can close the coincidence problem.

7.2. **Retired coincidence routes.** (i) Cycle exclusion and displacement nonvanishing (Example 7.1). (ii) Mossé desubstitution descent: a recurrent SCC is closed under inflation and desubstitution and yields no smaller state. (iii) Pure algebra on $N_{\mathcal{C}}$: $N_{\mathcal{C}} = M$, $P_{\mathcal{C}} = I$ satisfies every identity of Theorem 5.2. (iv) The degree-two endpoint cone (Example 7.4). (v) Stacking further fixed-size ($|\mathcal{C}| = 3$) sieves: each is decidable for a given σ by constructing $\mathcal{B}_\sigma$ and adds proof value only with a uniform completeness statement. (vi) Requiring synchronization in *every* recurrent SCC: false in the corpus (60 exceptions, all escaping), and unnecessary by Theorem 5.1. (vii) The unsigned normalized-SCC transfer $L_{\mathcal{C}} N_{\mathcal{C}} = \Phi_3 L_{\mathcal{C}}$ asserted in earlier versions: incorrect when a child occurs reversed; the correct relation is $Q_3 S = \Phi_3 Q_3$.

8. Unresolved statements

For reference we collect the open statements with their exact form and dependencies.

(1) **G1** (Hypothesis 2.14): $\mathcal{B}_\sigma$ finite. Open for general $|\mathcal{A}| \geq 3$; known for $|\mathcal{A}| = 2$ [9, 19] and for the verified families; certified for each $\sigma \in \mathcal{S}$.
(2) **G1b-1** (Hypothesis 4.4): bounded discrepancy over reachable states. Open; asserted in unavailable notes; not used anywhere below.
(3) **G1b-2** (Open Problem 4.5): finitely many reachable irreducible states of bounded discrepancy. Open; with G1b-1 implies G1. Candidate mechanisms: Open Problem 4.15 (symbolic) and Conjecture 4.16 (geometric), the latter requiring also that reachable states be realized.
(4) **Seedwise bridge** (Open Problem 4.18): whether pure discrete spectrum forces termination from every seed, including illegal ones. Determines whether G1 is necessary for pure discrete spectrum.
(5) **SCC Producer** (Conjecture 2.15); under G1 equivalent to nonexistence of strict components (Theorem 5.1).
(6) **Concentration** (Open Problem 5.22): no strict component with $K_2 \equiv 0$ ($d = 3$). By Proposition 5.20 this is the "concentration / aux-B" step of the spectral route, and the "span-rich" step then follows from it. Open.
(7) **Wedge productivity** (Open Problem 5.23): no strict component with $K_2 \not\equiv 0$ ($d = 3$). Open; equivalent to step (c) of the spectral route.
(8) **C4 / C3-local** (Open Problem 5.29): every strict component has a one-step interior synchronizing zero return. Implies SCC Producer under G1 (Proposition 5.8).
(9) **Higher first defects**: no proof that $r(\mathcal{C}) \leq 3$ for strict components (this is part of Open Problem 5.22); Theorem 5.17 restricts but does not exclude $r \geq 5$, nor $r = 4$ in the unimodular Perron-extremal case.
(10) **Realization** (Section 5.9): identification of coincidence rank > 1 with a globally realized producer-free component; and an effective bound on the collar radius at which unrealizable formal cycles die. Both open; the second is what would turn collar experiments into certificates.

9. Discussion

The balanced-pair route converts the Pisot substitution conjecture into two statements about a finite graph, and this paper has tried to say exactly what is known about each. On the finiteness side the situation is uncomfortable: unique decodability is a five-line consequence of $\det M \neq 0$ and is irrelevant to finiteness; the natural bounded-norm statements are either open (Hypothesis 4.4) or insufficient (Open Problem 4.5); and the only uniform finiteness theorem we can prove

(Theorem 4.13) concerns ancestry defects, which are known to admit nonzero cycles. The finite computations show what a proof must do, namely control which labelled return words are realized by the substitutive hierarchy across all levels, and they show that bounded windows of any fixed size cannot do it. The geometric twin of G1, finiteness of realized overlaps, is a theorem in the overlap algorithm [35, 2, 21]; the gap between the two is the relative density of simultaneous boundaries (Conjecture 4.16) together with the realization of formal states, and we think this is where a proof of G1b-2 should be sought, in an internal space that includes the non-Archimedean factor when $|\det M| > 1$ [26].

On the coincidence side the reductions are substantial but do not close: a strict component must be a no-leakage block substitution at the Perron rate, with full Parikh rank and at least $|\mathcal{A}|$ states, with both endpoint maps non-synchronizing and (for three letters) not 3-cycles, with a signed derived incidence of spectral radius at most β, with a first defect lying in a free Lie component subject to the sieves of Theorems 5.16–5.17, and with locally recognizable hierarchies on both sides. Two independent-looking routes, the spectral one and the boundary one, funnel into the same question: the spectral route splits it by first defect degree into Open Problems 5.22 and 5.23, the boundary route poses it as Open Problem 5.29; we regard the identification of the concentration step with the higher-degree case as a clarification, not as progress, and we keep it in the list of unresolved statements. What is missing is an argument that uses the actual ordered factorization of inflated states, not an invariant of it; the synthetic template of Section 7.1 shows how much a hypothetical component can satisfy before it fails, and it fails only at the factorization itself.

We end by restating the status. The Pisot substitution conjecture is open. Hypothesis G1 is open for $|\mathcal{A}| \geq 3$, and its remaining content is Open Problem 4.5, which is open. Conjecture 2.15 is open. Every computation reported here is finite and is labelled as such. Theorem 3.1 is a conditional statement, and the conditions are the whole difficulty.

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